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Research Project Periodic Log Entry Form

Project ID: R26-DS-010

Student ID: IT22178640

Name: Dayana Priyadharshani Kumar

Individual Component: AI Assisted Caregiver Workforce Deterioration Detection and Adaptive Workload Redistribution

Summary of RP work carried out during the reporting period

Date	RP Work
01.04.2026	I initialized the shared Git repository, added all team members as collaborators, and created a clear README file outlining all four components. I also used the ClickUp platform to assign and distribute tasks among the team. Additionally, I critically reviewed two research papers to identify suitable models that I can use for comparison when developing my own model to detect deterioration.
03.04.2026	Conducted a focused literature review on clinical nursing workload measurement methods by analyzing established research studies and industry standards. Based on these findings, I mapped relevant workload indicators such as workload intensity, schedule stability, and environmental strain to our available data architecture to ensure alignment between theoretical models and our dataset structure.
04.04.2026	Conducted an in-depth evaluation of open-source clinical and physiological datasets by reviewing State-of-the-Art (SOTA) sources used in stress detection and healthcare workload modeling. I specifically analyzed datasets such as the UCI Machine Learning Repository WESAD dataset and the SWELL Project dataset to assess their relevance to my research objectives.
05.06.2026	Critically assessed the limitations of the selected datasets and made informed decisions on their suitability. I rejected the WESAD dataset as it focuses only on short-term and lab-induced stress (Trier Social Stress Test) and does not capture long-duration physical exertion, patient-related workload. After EDA, I also rejected the SWELL dataset since it is limited to sedentary office environments and lacks



	key features such as physical movement data and environmental strain metrics relevant to healthcare settings. Based on this evaluation, I justified the need for novel data collection.
07.04.2026	Conducted a group interview with Mr. Vidyantha Vasantha Abeyrathne, the founder of Remark Elder Care (a subsidiary of Feathered Health Care), and visited the care home to gather real-world caregiver requirements. Based on the insights and meeting notes, I designed a workflow to predict caregiver stress using periodic voice logs and proposed a solution for workload redistribution through substitute caregivers.
08.04.2026	We, as a group had a meeting with the supervisor to discuss and refine my component based on the previous day's discussion, and clarified key points. I also reviewed the medical transcription dataset (mtsamples.com) available on Kaggle, but after performing exploratory data analysis, I found it unsuitable for my component as it mainly contained static clinical text records and lacked relevant features such as behavioral patterns, stress indicators, and temporal data needed for modeling caregiver deterioration.
10.04.2026	Conducted a literature review on the USC Institute for Creative Technologies DAIC-WOZ dataset. I analyzed its structure, which includes 189 clinical interview sessions with transcripts, audio recordings, and facial feature data aimed at detecting psychological distress such as anxiety and depression. However, I found that the dataset primarily focuses on facial expressions, and the audio components were not sufficiently aligned with my requirement for voice-based stress analysis. Therefore, I decided to move forward with other literature sources more suitable for my component.
11.04.2026	Analyzed the StressID dataset from the EURECOM repository, which is primarily structured using Python and Jupyter Notebooks. I observed that instead of a simple binary classification (stressed vs. normal), the dataset provides ground truth through four continuous self-assessment metrics. However, I identified a key limitation in using StressID for caregiver deterioration, as the stress is induced through controlled stimuli that do not reflect real-world caregiving conditions. I then proceeded to analyze the IEMOCAP dataset to further explore its suitability for emotion and stress detection in more natural interaction settings.
12.04.2026	Assessed the suitability of the MIMIC-IV dataset for my component by exploring how it can be used to infer shift times, calculate staffing ratios, and quantify caregiver workload. I specifically analyzed metrics such as event density per hour to understand workload patterns. Additionally, I initiated the required credentialing process to access the dataset, which includes CITI training, and have currently completed 1/4 of the training

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13.04.2026	Submitted the credentialing application to access the MIMIC-IV, which is currently under review. Following this, I completed the CITI -Conflicts of Interest Course, which focuses on identifying, managing, and disclosing potential conflicts in research. This training strengthened my understanding of ethical responsibilities and transparency requirements, and I also added it to my LinkedIn profile as a license to support the approval process. In parallel, I accessed and explored the available demo dataset to understand its structure and assess its potential for analyzing caregiver workload and related metrics.

Supervisor's Comments

Student progressing well in research project: Yes / No
If No:

Project work to be carried out before next evaluation period:

A project with a good progress.
After the proposal presentation, quickly moved to the next stages of the project and updated me.

Code base checked: Yes/No

Signature:

Date: 2026-04-14

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Research Project Periodic Log Entry Form

Project ID: R26-DS-010

Student ID: IT22057488

Name: Harishalinee Elangovan

Individual Component: Edge-Enabled Skeletal Movement Analysis System for Real-Time Patient Observable Fall Risk Detection and Prediction

Summary of RP work carried out during the reporting period

Date	RP Work
01.04.2026	Conducted a comparative analysis of available human action recognition datasets to support the Spatio-Temporal Graph Convolutional Network (ST-GCN) training. Identified the **NTU RGB+D Action Recognition Dataset** for skeletal movement modeling and the **UR Fall Detection Dataset** for environment-specific fall validation as the most critical sources for the component's architecture.
02.04.2026	Drafted and forwarded a formal request to the supervisor, outlining the technical justification for using the NTU RGB+D and UR Fall datasets. Requested official approval to utilize the supervisor's institutional credentials, which are strictly required by the dataset portals to authorize access for student research purposes.
04.04.2026	Initiated the setting up in the local edge computing environment. Installed and configured the necessary dependencies and lightweight real-time pose estimation frameworks.
07.04.2026	Conducted a field visit to the Remark Elder Home with the project team. Analyzed typical room dimensions, furniture placement, and real-world patient movement patterns. This observational data is crucial for determining optimal camera placement angles and understanding potential occlusion challenges for the pose estimation model.
08.04.2026	Participated in a comprehensive progress meeting with the supervisor. Discussed post-viva architectural improvements, specifically detailing how the system will transition from binary fall detection to a continuous progressive risk score (normal, moderate, high risk) using contextual motion features like body orientation and velocity.

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09.04.2026	Received formal authorization from the supervisor regarding dataset access. Successfully compiled the required documentation, including screenshots of the selection details and project scope, and officially submitted the data request forms to the respective institutional repositories for NTU RGB+D and UR Fall Detection.
11.04.2026	While awaiting final access to the requested bulk datasets, I began developing the foundational preprocessing pipeline using sample data. Engineered the script to parse raw video, extract the skeletal stick-figure representation, and immediately discard the original RGB visual frame from memory to enforce the system's core privacy mandate. ✓
13.04.2026	Conducted a detailed analysis of the skeletal coordinate data output generated by the initial preprocessing pipeline. Reviewed the data structure (e.g., JSON/array formats) to understand how the extracted joints are represented. Additionally, began researching PyTorch-based implementations of Spatio-Temporal Graph Convolutional Networks (ST-GCN) via open-source academic repositories to understand the required input dimensions and mathematical formatting needed for the upcoming model development phase.

Supervisor's Comments

Student progressing well in research project: Yes / No
If No:

Project work to be carried out before next evaluation period:

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Code base checked: Yes/No

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Research Project Periodic Log Entry Form

Project ID: R26-DS-010

Student ID: IT22294098

Name: Vidursha Prabagaran

Individual Component: Context-Aware Intelligent System for Automated ADL Extraction and Real-Time Documentation from Caregiver Voice Logs

Summary of RP work carried out during the reporting period

Date	RP Work
01.04.2026	Focused on gaining a clear understanding of the research component in detail. Carefully studied the objectives and scope of the component. Collected and reviewed several research papers related to the component to understand existing methods and technologies used in similar systems.
02.04.2026	Worked on combining all previously proposed features into one complete system structure design. First, sketched the full system design including all planned features. Each feature was carefully analyzed to evaluate its feasibility for implementation. Some parts were modified based on the feasibility analysis, and the complete component diagram was re-sketched with all finalized features.
04.04.2026	Conducted a detailed literature review related to Activities of Daily Living (ADL) extraction, which is one of the main features of the component. Studied multiple research papers to understand how ADL extraction is performed in similar systems.
05.04.2026	Explored the concept of context reasoning while reviewing research papers. Identified that Transformer-based models are commonly used for this task. After analyzing several studies, determined that BERT and BioBERT models are suitable for this component. Based on these findings, one of these models will be selected as an appropriate choice for implementation.
06.04.2026	Participated in a discussion with an experienced caregiver who has more than 30 years of experience working in four different countries. Gained detailed knowledge about Activities of Daily Living (ADL) and how caregivers perform daily care activities in real caregiving environments. Also learned about applications used in



	caregiving systems in different countries. Detailed notes were recorded during this discussion.
07.04.2026	Participated in a field visit to Remark Elder Home with the project team. Observed how daily caregiver logs are currently recorded. Gained an understanding of the challenges faced in maintaining caregiver records and the systems currently in use. The limitations of existing systems were also explained during the visit. This visit provided valuable insight into real-world challenges and documentation needs.
08.04.2026	Explored existing caregiver applications used for recording care logs and documentation. Analyzed how these applications function in real caregiving environments. Identified common issues in these systems and took notes on possible improvements for the component. Also had a meeting with the supervisor to discuss findings and ideas.
09.04.2026	Focused on identifying suitable datasets for ADL data collection in caregiving environments. Several available datasets were reviewed to determine their relevance and suitability for the research component. Based on this review, the MIMIC-IV dataset from PhysioNet and the Activities of Daily Living (ADL) Dataset from IEEE DataPort were selected for further use. A request was also submitted to obtain access to the ADL dataset available on IEEE DataPort.
11.04.2026	Focused on studying data collection methods used in research related to caregiving systems. Carefully reviewed the methodology sections of several research papers. Analyzed how data was collected in those studies and identified suitable data collection approaches that can be applied to my research component.
13.04.2026	Focused on identifying the specific data required for my research component. Reviewed several research papers and datasets to understand the types of data commonly used in similar systems. Based on this analysis, finalized the types of data that need to be collected for my research component and identified the features that need to be extracted. Created a clear list of required data elements that will be used in the next stages of the research.

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Research Project Periodic Log Entry Form

Project ID: R26-DS-010

Student ID: IT22289384

Name: Kaushalya Nagenthraraja

Individual Component: Personalized Sign Language Tutoring System for Caregivers

Summary of RP work carried out during the reporting period

Date	RP Work
01.04.2026	Reviewed key research in sign language technology and education to further understand current system capabilities and learning gaps. Identified that existing approaches mainly emphasize recognition accuracy through methods such as CNNs, HMMs, and skeleton-based models. Also observed that effective learning systems should go beyond simple correct/incorrect outputs, incorporating feedback with strong emphasis on visual guidance to support error correction and skill improvement.
02.04.2026	I continued to compile caregiver-focused sign language vocabulary using insights from literature and domain understanding. I also identified key categories such as Activities of Daily Living (eating, drinking) emergency and pain-related communication (pain location, fever), comfort and emotional expressions, as well as basic instructional signs. I also explored healthcare communication studies to better understand the most critical interaction points between caregivers and patients. As part of this ongoing work, an initial list of approximately 30 to 40 signs was listed.
03.04.2026	I reviewed two relevant healthcare communication studies to validate, I noted common breakdowns in areas such as symptom reporting and emotional reassurance, where written communication is insufficient and sign language becomes essential. I also observed that while professional interpreters are ideal, their limited availability highlights the importance of equipping caregivers with fundamental sign language skills to bridge communication gaps.

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04.04.2026	Conducted a comprehensive evaluation of open-source datasets relevant to sign language gesture research, including the ASL Citizen dataset and the WLASL video dataset, exploring how they align with my research objectives. I primarily researched datasets on Kaggle and explored other open-source platforms such as GitHub to identify additional relevant resources for sign language data.
06.04.2026	I prepared a set of structured questions and noted key ideas to clarify during the interview with the COO of Remark Elder Care Home. I organized both component-wise and general questions to support individual and group research needs and ensure a clear understanding of caregiving practices and requirements.
07.04.2026	Visited the Remark Elder Care Home in Battaramulla and conducted an interview with the COO to gain an in-depth understanding of caregiving practices and the live-in care model in Sri Lanka. He also shared his perspectives and suggestions relevant to my research, emphasizing the importance of integrating non-verbal sign-based communication for conditions such as Alzheimer's, dementia, and depression, and highlighted the need to educate caregivers on these specialized signs to improve patient care.
08.04.2026	I conducted a broad analysis of common conditions in elder care, including Parkinson's, Alzheimer's, Dementia, and Cancer, with a focus on identifying associated facial expressions, postural changes, and behavioral patterns. After evaluation, I narrowed the scope by excluding Parkinson's and Cancer, and focused on Alzheimer's, Dementia, and late-life Depression related to overlapping non-verbal sign vitals such as confusion, gaze aversion, psychomotor slowing, and wandering behavior.
09.04.2026	Dedicated the day to validating the non - verbal sign vitals identified earlier against academic literature and cross-referencing them with observational notes from the Remark Elder Care Home visit. I confirmed the consistency between clinical research and real-world behavioral patterns observed in patients. Reviewed papers on depression-related visual cues such as flattened affect, reduced facial expression, and gaze aversion and video-based behavioral analysis in elder care, highlighting dementia-related patterns like wandering and agitation.
11.04.2026	I moved into the implementation phase by evaluating the optimal tech stack for real-time spatiotemporal analysis. I selected OpenCV and MediaPipe for facial landmark detection and pose estimation, alongside Python for backend processing of behavioral patterns.

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12.04.2026	Tried to set up the development environment and experimented with MediaPipe by running a basic real-time landmark detection example using a webcam. I was able to observed facial/hand landmark tracking and gained initial familiarity with how the framework detects and visualizes key points.

Supervisor's Comments

Student progressing well in research project: Yes / No
If No:

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Project ID: R26-DS-010

Student ID: IT22178640

Name: Dayana Priyadharshani Kumar

Individual Component: AI Assisted Caregiver Workforce Deterioration Detection and Adaptive Workload Redistribution

Summary of RP work carried out during the reporting period

Date	RP Work
15.04.2026	Explored a multimodal approach like physiological signals such as heart rate and Electrodermal Activity (EDA). A suitable Kaggle dataset was selected and preprocessed through cleaning, normalization, and feature structuring. Supervised machine learning models including Logistic Regression and Random Forest were trained using the labeled physiological data. While the models showed reasonable performance, several limitations were identified. The dataset was relatively constrained and may not fully represent real-world caregiver conditions, affecting generalizability. Additionally, physiological data can be sensitive to external factors such as movement, environment, or device inconsistencies, which may introduce noise.
17.04.2026	Further analysis was conducted using unlabeled data with unsupervised learning techniques such as K-Means, DBSCAN, and Agglomerative Clustering to identify hidden patterns. During multimodal evaluation, it was observed that voice data, particularly environmental noise, significantly distorted feature extraction and falsely amplified stress indicators. This made environmental voice a dominant but unreliable factor, reducing overall model effectiveness. Based on these findings, a refined approach was decided: instead of relying on physiological input, the system will incorporate structured daily voice summaries focusing on tone, along with shift/workload details as contextual data, providing a more reliable and practical solution for stress detection.
18.04.2026	Reviewed research papers focusing on stress detection using speech signals. One key paper, *"An automatic speech analytics program for digital assessment of stress burden and psychosocial health"* demonstrated that speech analysis can identify caregiver stress levels with around 72% accuracy, showing that voice can be used as a non-invasive and scalable



	method for early stress detection. Key findings and the limitations found were noted down for the future checklist creation.
19.04.2026	Reviewed another study, *"Stress Detection Through Speech Analysis Using Machine Learning"*, it has mentioned that they have used MFCC feature extraction and deep learning models (CNN) to classify stress from audio signals. It highlighted the importance of spectrogram-based analysis and acoustic features such as pitch and energy. Additionally, research paper, *"VoStress– Voice-based Detection of Acute Psychosocial Stress"* emphasized that voice can serve as an alternative to traditional stress measurement methods, which are often expensive and time-consuming.
21.04.2026	Visited CCHS (Colombo Caregiver & Health Science School) to gain a practical understanding of real-world caregiving workflows and documentation practices. The purpose of the visit was to observe how caregivers are trained, how they manage daily responsibilities, and how they record patient-related information. Special attention was given to understanding the logging process maintained by students, particularly how they document the condition, behavior, and daily care activities of the individuals they are assigned to.
22.04.2026	Visited a local elder care home *"Love and Care Home"* on two occasions to closely observe real world caregiving practices and documentation methods. The visits focused on understanding how caregivers manage different types of patients and how daily logs are maintained in practical settings. Interactions were held with the director and caregiving staff to gain deeper insights into operational workflows and patient-specific care routines. It was observed that log records vary significantly depending on patient conditions; for example, kidney patients require detailed monitoring of fluid intake, output levels, and dietary restrictions, while other patients may require logs related to mobility, medication schedules, or behavioral changes.
24.04.2026 and 25.04.2026	Collected a set of voice recordings from approximately 10–15 individuals simulating caregiver daily summaries to analyze speech patterns, tone variations, and content structure. Each recording included a brief verbal log describing patient condition, tasks performed, and observations. These recordings were then transcribed and summarized to compare spoken content with structured log formats. Additionally, tone-related aspects such as pitch, speaking rate, and pauses were analyzed to identify potential stress indicators. This task helped in understanding the variability in voice inputs and highlighted the challenges of extracting consistent features for stress detection, especially due to differences in speaking style and background noise.
28.04.2026	Conducted further analysis on the collected voice recordings by organizing them into labeled categories based on perceived stress levels (e.g., low, moderate, high) using manual observation and tone interpretation. Performed basic preprocessing steps such as noise reduction and normalization to improve audio quality.

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Supervisor's Comments

Student progressing well in research project: Yes / No If No:

Project work to be carried out before next evaluation period:

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Date 2026-04-29



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Research Project Periodic Log Entry Form

Project ID: R26-DS-010

Student ID: IT22057488

Name: Harishalinee Elangovan

Individual Component: Edge-Enabled Skeletal Movement Analysis System for Real-Time Patient Observable Fall Risk Detection and Prediction

Summary of RP work carried out during the reporting period

Date	RP Work
14.04.2026	Analyzed feedback received during the proposal presentation and identified key areas requiring improvement, including the need to transition from binary fall detection to a progressive risk-based approach. The system design revised to incorporate three risk levels (normal, moderate, high) supported by temporal movement analysis.
15.04.2026	Refined the overall system architecture by defining a structured pipeline that transforms raw video input into skeletal representations and subsequently into temporal feature sequences. Established a modular design to support future integration between preprocessing, modeling, and interface components.
16.04.2026	Organized the UR Fall Detection dataset into a structured format suitable for processing. Segregated fall and ADL sequences and standardized directory structures to support automated preprocessing workflows.
17.04.2026	Implemented automated preprocessing routines to manage dataset extraction and restructuring. Addressed inconsistencies in directory hierarchy and ensured uniform access patterns for large-scale sequence processing.



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18.04.2026	Conducted a detailed evaluation of pose estimation frameworks with a focus on real-time applicability and computational efficiency. Selected MediaPipe Pose as the primary framework due to its suitability for edge-based deployment and ability to generate reliable skeletal joint coordinates.
19.04.2026	Designed and implemented the skeletal extraction pipeline to convert RGB image sequences into structured joint coordinate data. Ensured temporal consistency across frames and maintained sequence-level organization for downstream analysis.
20.04.2026	Introduced privacy-preserving mechanisms into the preprocessing pipeline by ensuring that original RGB frames are discarded immediately after skeletal extraction. This design enforces anonymization and aligns with ethical requirements for patient monitoring systems.
21.04.2026	Conducted a field visit to Love Care Elder Home to observe real-world environmental conditions and patient movement patterns. Identified key fall-risk scenarios and spatial constraints, contributing to improved system assumptions regarding camera placement and movement variability.
22.04.2026	Visited Colombo Caregiver & Health Science School to gain domain knowledge on caregiving practices and fall prevention. Engaged with professionals to understand common instability indicators and incorporated these insights into system requirements.
23.04.2026	Enhanced the robustness of the skeletal extraction pipeline by implementing mechanisms to handle missing or low-confidence detections. Ensured continuity in temporal sequences by introducing fallback strategies for incomplete frames.
24.04.2026	Initiated processing of the NTU RGB+D dataset by analysing the structure of skeletal data files. Examined joint indexing and temporal segmentation to align the dataset with the existing preprocessing framework.
25.04.2026	Developed parsing methods to extract joint coordinate sequences from NTU skeletal data. Standardized the output format to ensure compatibility with the UR dataset, enabling unified data representation.
26.04.2026	Defined the classification strategy for fall risk prediction by categorizing movement patterns into normal, moderate instability, and high-risk fall conditions. Established label mappings based on dataset characteristics and research objectives.
27.04.2026	Planned the integration of UR and NTU datasets into a single training pipeline. Structured the data flow for combining multiple sources and prepared the framework for model training and evaluation.

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28.04.2026	Designed the initial prototype workflow, outlining the interaction between preprocessing, model prediction, and user interface components. Defined core functionalities including skeletal extraction and fall risk prediction and prepared the system for implementation of a basic model to demonstrate proof of concept.

Supervisor's Comments

Student progressing well in research project: Yes / No If No:

Project work to be carried out before next evaluation period:

Code base checked: Yes/No

Signature:

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Project ID: R26-DS-010

Student ID: IT22294098

Name: Vidursha Prabagaran

Individual Component: Context-Aware Intelligent System for Automated ADL Extraction and Real-Time Documentation from Caregiver Voice Logs

Summary of RP work carried out during the reporting period

Date	RP Work
16.04.2026	Focused on exploring the MIMIC-IV dataset to evaluate its relevance to my research component. Identified that the dataset contains a large amount of Intensive Care Unit (ICU) data, which is highly clinical in nature. ICU (Intensive Care Unit) refers to a specialized hospital department where critically ill patients are closely monitored and treated. Through this exploration, developed a clear understanding of how structured medical records are organized and maintained in large-scale healthcare datasets. This knowledge will support the design of the data structure for my research component.Additionally, initiated the required credentialing process to gain access to the dataset through PhysioNet. As part of this process, began the CITI training program, focusing on the Conflicts of Interest course and the Data or Specimens Only Research course. Successfully completed the Conflicts of Interest course and made progress in the Data or Specimens Only Research course by completing 4 out of its 9 modules.
17.04.2026	I successfully completed the Data or Specimens Only Research course as part of my CITI training requirements. I also submitted both the Conflicts of Interest course and the Data or Specimens Only Research course. These courses taught me how to recognize, manage, and properly declare any conflicts that may happen during research. This training helped me improve my knowledge of research ethics, correct data handling methods, and the importance of being honest, transparent, and responsible when carrying out research work.



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19.04.2026	Carefully reviewed the methodology sections of two research papers related to ADL extraction and context-aware systems. Analyzed how similar studies were designed, including their data collection techniques, preprocessing methods, and model implementation strategies. This review helped in refining my own research methodology and provided insights into best practices that can be applied to improve the quality and reliability of my research component.
20.04.2026	Prepared detailed notes prior to visiting Love and Care Elder Home, outlining the key discussion points and identifying the specific types of data required for my research component. Conducted a field visit to Love and Care Elder Home, where initial observations were made regarding caregiving practices and documentation processes. Engaged in discussions with staff to understand how daily caregiver logs are maintained. Successfully requested access to relevant data, and the management agreed to support and provide data for my research component.
21.04.2026	Visited CCHSS - Colombo Caregiver & Health Science School to gain a better understanding of real-world caregiving practices. During the visit, I observed how caregivers are trained and how their daily care activities are organized and recorded. I carefully focused on how daily caregiver logs are maintained in a professional setting. I also learned that caregivers do not need very deep medical knowledge. instead, they mainly require basic medical knowledge to carry out their duties safely. Additionally, I was given a caregiver daily work guidebook, which I reviewed to understand standard caregiving procedures and documentation formats. This visit helped me identify practical data requirements for my research.
22.04.2026	Focused on planning the data collection process for ADL (Activities of Daily Living) logs at Love and Care Elder Home. Since manual data collection was required, researched and identified effective approaches for collecting, organizing, and preparing data. Also explored suitable tools and technologies that can be used for recording, managing, and preprocessing the collected data. This preparation ensured readiness for the upcoming data collection phase.
26.04.2026	Conducted a full-day field visit to Love and Care Elder Home, staying from morning until night to observe and collect real caregiving data. Initiated the data collection process by gathering daily caregiver logs for 18 female elderly patients. Data was collected through direct observation as well as by obtaining existing caregiver records. Detailed notes were taken, and the collected information was carefully documented in written form to ensure accuracy and completeness. This hands-on



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	data collection provided valuable real-world insights and essential data for the development of the research component.
27.04.2026	Visited another branch of Love and Care Elder Home and conducted a full-day data collection session from morning to night. Collected daily caregiver logs for 21 male elderly patients through direct observation as well as by obtaining existing caregiver records. Detailed notes were taken, and the collected information was carefully documented in written form to ensure accuracy and completeness. This expanded dataset significantly contributed to building a more diverse and comprehensive dataset for the research component.

Supervisor's Comments

Student progressing well in research project: Yes / No If No:

Project work to be carried out before next evaluation period:

Code base checked: Yes/No

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Date 2026-04-29



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Research Project Periodic Log Entry Form

Project ID: R26-DS-010

Student ID: IT22289384

Name: Kaushalya Nagenthraraja

Individual Component: Personalized Sign Language Tutoring System for Caregivers

Summary of RP work carried out during the reporting period

Date	RP Work
16.04.2026	Finalized the caregiver-oriented sign language vocabulary list following a detailed review of the WLASL dataset structure and relevant medical sign language resources. A total of 50 essential signs were compiled and systematically categorized into four key groups: Daily Care (15 signs, including eat, drink, medicine, bathroom, sleep), Emergency Communication (10 signs such as help, pain, fall, doctor, ambulance), Emotional Support (12 signs including comfort, sad, happy, calm, scared), and Basic Instructions (13 signs such as sit, stand, walk, come, stop).
17.04.2026	Attempted to implement basic hand landmark detection using the MediaPipe Hands solution. During testing, I observed issues with tracking stability, particularly when hands moved quickly or partially moved out of the camera frame. The system frequently lost track of hand landmarks during fast signing movements, which is critical for accurate sign language recognition.
18.04.2026	Explored alternative approaches to improve hand tracking stability using MediaPipe Hands and MediaPipe Holistic. Conducted testing to compare both models, observing that Holistic provided more stable tracking by incorporating pose information. However, it operated at a lower frame rate, highlighting a trade-off between stability and real-time performance.
19.04.2026	Conducted experiments to optimize MediaPipe performance under different conditions. Tested multiple webcam resolutions (640×480, 1280×720, 1920×1080) and observed that 1280×720 provided a good balance between accuracy and processing speed. Also evaluated the impact of lighting and found that stable, frontal lighting significantly improved hand landmark detection, while poor lighting conditions (e.g., backlighting) reduced accuracy. Based on these tests, defined basic



	recording guidelines including proper lighting, clear background, and appropriate camera distance.
20.04.2026	Attempted to download the WLASL dataset using the official GitHub video downloader script (video_downloader.py). During execution, encountered multiple issues due to broken or unavailable YouTube links. As a result, only around 60% of the required videos were retrieved. Explored alternative sources and identified a preprocessed WLASL subset on Kaggle, which was successfully downloaded for further use. To address this limitation, explored alternative data sources and identified a preprocessed WLASL subset available on Kaggle. Proceeded to download the WLASL-100 subset, which contains 2,038 labeled video samples across 100 commonly used signs. This version provided a more stable and ready-to-use dataset for initial model development and experimentation.
21.04.2026	Visited the Love and Care Elder Home to understand real-world caregiving environments and communication challenges. Observed interactions between caregivers and elderly residents, particularly those with dementia and limited verbal ability. Engaged with caregivers to identify common difficulties in expressing basic needs and emotions. Also observed non-verbal sign vitals related to dementia, Alzheimer's, and Parkinson's disease, including reduced facial expressions, slow movements, and confusion.
22.04.2026	Visited CCHS, a caregiver training association, to understand how formal training is conducted and to identify gaps between training practices and real-world caregiving scenarios. During the visit, also gained insights into commonly used caregiving vocabulary, which can be adapted for sign language development.
24.04.2026	Compared the previously defined caregiver-focused vocabulary with the WLASL-100 dataset to assess coverage. Identified 18 matching signs (e.g., help, eat, drink, sleep, sit, stand, walk, stop, happy, sad, pain, medicine, bathroom, doctor, come, fall, calm, scared), confirming partial alignment with available labeled data. For the remaining signs not present in WLASL-100, considered two approaches: searching within the full WLASL dataset and exploring the possibility of collecting custom video samples for missing caregiver-specific signs.
25.04.2026	Built on earlier observations from the elder care home visits and identified non-verbal communication cues, I further researched techniques for detecting non-verbal sign vitals related to conditions such as Alzheimer's, dementia, and depression. Reviewed academic approaches on facial expression and behavior analysis, focusing on measurable features like blink rate, gaze patterns, and head movements.
26.04.2026	As part of implementation, developed a basic proof-of-concept using MediaPipe Face Mesh to track facial landmarks in real time. Implemented blink detection using the Eye Aspect Ratio (EAR) method and tested it through a webcam feed. Initial results showed reliable blink



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	detection, supporting the feasibility of using facial features for non-verbal sign vital analysis in the system.
27.04.2026	Started implementing a basic sign recognition pipeline using the WLASL-100 dataset. Extracted hand landmark features using MediaPipe Hands but noticed inconsistencies in some frames due to unstable tracking.
28.04.2026	Looked into basic preprocessing steps such as frame normalization and handling variable-length video sequences. Faced challenges in maintaining consistent sequence data due to differences in video lengths and occasional missing landmark values, indicating the need for more refined preprocessing techniques.
29.04.2026	Initiated development of a temporal model using Long Short-Term Memory (LSTM) networks to better capture motion patterns. Faced challenges in preparing consistent sequence data, as video lengths varied and some frames had missing landmarks, requiring additional preprocessing and padding strategies.

Supervisor's Comments

Student progressing well in research project: Yes / No If No:

Project work to be carried out before next evaluation period:

Code base checked: Yes/No

Signature:

Date 2026-04-29



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Research Project Periodic Log Entry Form

Project ID: R26-DS-010

Student ID: IT22178640

Name: Dayana Priyadharshani Kumar

Individual Component: AI Assisted Caregiver Workforce Deterioration Detection and Adaptive Workload Redistribution

Summary of RP work carried out during the reporting period

Date	RP Work
01.05.2026	Reviewed and finalized dataset selection rationale for the Data Analysis Report. Confirmed Multimodal Occupational Stress Recognition Dataset (Hosseini & Akhavian, 2022) as primary physiological training source. Documented IBI.csv discovery and all 529 sessions returned empty IBI files, resolved by implementing HRV extraction from raw 64Hz BVP signal using NeuroKit2 PPG peak detection. Updated preprocessing pipeline self notes accordingly.
02.05.2026	Completed BVP startup artifact removal implementation and first 10 seconds of each session discarded to eliminate Empatica E4 optical sensor stabilization transients. Verified ACC-EDA correlation feature (eda_acc_corr) correctly flags movement-contaminated EDA peaks. Ran full feature extraction pipeline on all 15 nurses producing final feature matrix: 13,287 windows × 30 features. Confirmed label distribution: No Stress 18.8%, Medium Stress 6.6%, High Stress 74.5%
03.05.2026	Completed physiological model training in Google Colab Notebook 2. Ran three-model LOSO comparison: Logistic Regression Binary F1 = 0.551, Random Forest Binary F1 = 0.822, XGBoost Binary F1 = 0.816. Documented XGBoost selection rationale and SHAP TreeExplainer requires XGBoost-format trees for exact Shapley computation; RF SHAP is approximated and non-deterministic. Trained production model on all 15 nurses: Binary F1 = 0.861. Saved physio_model_binary.pkl (667 KB) and scaler_binary.pkl to Google Drive.



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04.05.2026	Ran SHAP TreeExplainer on all 15 nurses using held-out LOSO windows. Computed per-nurse top-5 feature profiles. Key finding: 10 of 15 nurses have eda_peaks_count as dominant stress biomarker (electrodermal responders); Nurses 5C, 6B, 6D have temp_mean dominant (thermoregulatory responders); Nurses CE, DF have eda_mean dominant. Saved shap_by_nurse.json (11 KB). Confirmed this addresses the gap identified in Hosseini & Akhavian (2022) and they report only global feature importance.
05.05.2026	Completed Isolation Forest personalized baseline ablation study. Fitted IsolationForest (contamination=0.1, n_estimators=100) per nurse on 7-day calibration window. Compared three strategies: fixed threshold FPR = 41.2%, population baseline FPR = 28.7%, Isolation Forest FPR = 18.3%. Saved ablation_comparison.csv and 15 isolation forest .pkl model files to Google Drive Processed/ folder. Documented as empirical answer to panel proposal comment on individual behavioral pattern identification.
06.05.2026	Completed audio model pipeline in Google Colab Notebook 3. Rejected MIRSD dataset (lexical stress — F1 = 0.483, near-chance) and switched to RAVDESS. Implemented 34 language-independent acoustic features using librosa. Ran 5-fold actor-independent cross-validation comparing LR (F1=0.780), RF (F1=0.770), XGBoost (F1=0.761). Trained 1D CNN (Conv1D × 3 blocks, GlobalAveragePooling1D) achieving F1 = 0.804, AUC = 0.889. Saved audio_model_cnn.h5 (2 MB) and audio_model.pkl (399 KB).
07.05.2026	Completed late fusion and trajectory system in Google Colab Notebook 4. Computed performance-weighted fusion: physiological 51.7% + audio 48.3% (weights derived from validated F1 scores). Implemented daily risk score aggregation from LOSO window predictions. Computed slope via numpy.polyfit per nurse. Generated trajectory_data.json (24 KB) with all 15 nurses' daily_scores, future_scores, slope, mean_risk, alert_level, and days_to_threshold. Verified dashboard reads all values correctly.



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08.05.2026	Frontend integration of real model outputs into CareSense dashboard. Replaced all mock caregiver data with values from trajectory_data.json and shap_by_nurse.json across all 15 nurse cards and profile pages. Verified Nurse 5C critical trajectory (slope +2.73, score 88.9, delta +45.4), Nurse F5 stable (slope -0.58, delta 0), Nurse 6D recovering (slope -1.77, score 37.5). Confirmed SHAP bar charts render correct per-nurse features on each profile page.
09.05.2026	Updated modality donut chart to reflect real fusion weights (Physiological 51.7%, Audio 48.3%). Fixed trajectory chart to render correct n_days per nurse (F5 = 14 days, 7A = 33 days, 5C = 19 days) rather than hardcoded 14-day window. Updated Analytics & Trends page stat cards with real values: Physiological F1 = 86.1%, Audio F1 = 80.4%, 15 nurses, 13,287 signal windows. Updated network graph to show real nurse IDs (F5, 7A, 5C, 6B, 94 etc.) with real risk scores from model outputs.
10.05.2026	Finalized CareSense dashboard. Completed all five pages: Deterioration Detection main dashboard, 15 individual nurse profiles, Team Relationship Graph, Analytics and Trends, Adaptive Workload Redistribution. Conducted internal walkthrough to verify all values trace back to model outputs. Added Research Objective banners for Objectives 1, 2, 3, and 4 on respective dashboard sections. Updated Redistribution page nurse names from generic labels to real nurse IDs.
11.05.2026	Conducted pre-presentation walkthrough with supervisor to show PP1 progress and collect feedback before actual presentation. Addressed feedback: updated caregiver card badge colors for clearer critical/moderate/none distinction, added days-to-threshold display on trajectory section, confirmed SHAP chart rendering for all 15 nurses. Finalized PP1 presentation slides covering all four required sections: How the Solution Addresses the Sub-Problem, User Requirements, Design Excellence and Contributions (per-nurse SHAP, Isolation Forest ablation, 1D CNN application), and User Feedback on Prototype. Reviewed research paper master sheet and added additional papers with full details: Cho et al. (2023) JMIR nurse burnout AI mobile intervention, Rana et al. (2024) passive AI burnout frontline workers review, Mishra & Rizvi (2025) LSTM speech emotion detection, Dogan & Akbulut (2023) WorkStress3D multimodal fusion, Zhang & Liu (2022) Isolation Forest anomaly detection, Vaikole et al. (2020) CNN MFCC stress detection. Verified all paper

	entries include author, citation count, journal ranking, type, and detailed review notes.
12.05.2026	Delivered PP1 presentation. Presented individual component: AI-Assisted Caregiver Workforce Deterioration Detection and Adaptive Workload Redistribution. Demonstrated live CareSense dashboard including real model outputs for all 15 nurses, per-nurse SHAP charts, trajectory analysis, and Isolation Forest personalized baseline. Addressed panel questions on 3-class macro F1 (0.305), binary F1 (0.861), audio model (F1=0.804, AUC=0.889), XGBoost vs Random Forest selection, and individual behavioral pattern identification methodology. PP1 completed.
13.05.2026	Submitted PP1 deliverables. Submitted Master Sheet of Research Papers Studied and 19 papers covering caregiver stress detection, physiological signal processing, audio stress classification, multimodal fusion, personalized baseline modelling, temporal trajectory, graph-based workforce modelling, and clinical AI explainability. Submitted Data Analysis Report covering research background, dataset description and suitability analysis, complete preprocessing methodology, model training results (physiological and audio), SHAP attribution findings, trajectory analysis for all 15 nurses, and documented challenges with resolutions.
14.05.2026	Prepared TILES-2018 Dataset Data Use Agreement (DUA) for submission to USC Institute for Creative Technologies. Completed all required fields: research institution, study purpose, intended use of shift schedule data for graph edge construction (Objective 4) and longitudinal physiological data for Bi-LSTM temporal modelling (PP2). Requested supervisor signature on DUA form for institutional endorsement. TILES-2018 access required for PP2 milestones: graph-based relational workforce modelling with real shift co-worker edges, Bi-LSTM 10-week trajectory model training, and audio domain adaptation fine-tuning on real occupational speech features.

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Supervisor's Comments

Student progressing well in research project: Yes / No
If No:

Project work to be carried out before next evaluation period:

Code base checked: Yes / No

Signature:

Date: 2026/05/14



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Research Project Periodic Log Entry Form

Project ID: R26-DS-010

Student ID: IT22057488

Name: Harishalinee Elangovan

Individual Component: Edge-Enabled Skeletal Movement Analysis System for Real-Time Patient Observable Fall Risk Detection and Prediction

Summary of RP work carried out during the reporting period

Date	RP Work
29.04.2026	Conducted an extensive literature review utilizing Mendeley to systematically organize citations and extract methodologies for the Data Analysis Report. Focused specifically on identifying research gaps in cross-dataset domain shift mitigation and late-fusion architectures that combine classical biomechanical features with deep Spatio-Temporal Graph Convolutional Networks (ST-GCN).
30.04.2026	Designed high-fidelity UI wireframes for the React-based supervisor web dashboard. Outlined the structural layout for the 24-patient Dashboard overview, the Room-Overview bed map, and the Alerts/Risk tab to efficiently manage sustained moderate warnings and critical high-risk escalations without overwhelming the user.
01.05.2026	Created detailed UI wireframes for the caregiver Flutter mobile application. Emphasized the design of the explainability interface, specifically conceptualizing the privacy-preserving 5-second skeletal replay mechanism and the joint instability heatmap to provide caregivers with actionable, interpretable context.
02.05.2026	Investigated potential data leakage pathways from the Iteration 1 preprocessing pipeline. Created a secure backup repository (_backup_old_preprocessing/) to preserve the original random-stratified splits and Iteration-1 model outputs, ensuring a strict, transparent audit trail for before-and-after leakage proofing.
03.05.2026	Engineered Iteration-2 preprocessing fixes to address identified dataset leakage. Implemented spine-centered, torso-scaled spatial normalization to remove coordinate-system biases between the NTU and UR datasets. Additionally, replaced



	frame-freeze tail padding with linear interpolation to eliminate detectable static-tail artifacts.
04.05.2026	Executed Iteration-2 model retraining across the complete pipeline. Retrained the Random Forest baseline, the 4-class Posture Classifier, and the 3-million parameter ST-GCN deep model to ensure the networks learned genuine spatio-temporal motion dynamics rather than dataset-specific positional shortcuts.
05.05.2026	Developed and trained the Late-Fusion MLP model, combining a 24-dimensional vector of ST-GCN logits, RF probabilities, and 18 motion features. Evaluated the entire model suite under three escalating-difficulty protocols: Cross-Subject (subject-disjoint), Cross-View, and Cross-Dataset (NTU to UR).
06.05.2026	Compiled and analyzed the comprehensive multi-protocol evaluation results. Documented that the Fusion MLP achieved an optimal 94.24% test accuracy and an 87.43% high-risk recall in-domain, successfully reducing missed high-risk fall events by 42% compared to the ST-GCN standalone model.
07.05.2026	Initiated Phase 1 (PP1) backend-to-frontend data layer integration. Engineered the build_demo_predictions.py pipeline to aggregate real test-set predictions (Fusion softmax, posture labels, and 18 motion features) and assign 24 unique held-out test sequences to simulated patient slots.
08.05.2026	Finalized the backend generation of the static JSON manifest contract. Validated that the manifest correctly incorporated plain-English justification strings, action annotations, and dynamically rendered 14-joint skeletal MP4 animations for each of the 24 patient entries to power the frontend interface.
09.05.2026	Successfully wired the generated JSON manifest to the React dashboard via the demoFeed.ts loader. Verified that the PatientDetailPanel, DonutCharts, and Progressive Risk Indicators dynamically rendered the authentic trained-model outputs rather than seeded mock data.
10.05.2026	Successfully wired the generated JSON manifest to the React dashboard via the demoFeed.ts loader. Verified that the PatientDetailPanel, DonutCharts, and

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	Progressive Risk Indicators dynamically rendered the authentic trained-model outputs rather than seeded mock data.
11.05.2026	Conducted a comprehensive progress meeting with the supervisor. Demonstrated the fully integrated PP1 dashboard running live, showcasing the backend-to-frontend connection. Discussed the quantitative findings regarding the cross-dataset domain shift and how the Iteration-2 spatial normalization improved robustness.
12.05.2026	Delivered the formal Progress Presentation 1 (PP1) to the evaluation panel. Showcased the unified data preprocessing pipeline, the multi-protocol model evaluation strategy, and the fully integrated React dashboard. Received and documented panel feedback regarding future implementation steps.
13.05.2026	Finalized the system documentation based on the latest project achievements. Officially submitted the comprehensive Data Analysis Report, Progress Checklist 2, and the Data Analysts Master Sheet, ensuring all 12 model checkpoints and AI usage declarations were thoroughly detailed.
14.05.2026	Synthesized the feedback received from the PP1 evaluation panel. Began architectural planning for Phase 2 (PP2), specifically outlining the technical requirements to replace the static JSON manifest with a live Flask HTTP API and integrating the physical USB camera for real-time edge inference.

Supervisor's Comments

Student progressing well in research project: Yes / No
If No:

Project work to be carried out before next evaluation period:

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Code base checked: Yes/No

Signature: 

Date: 2026/05/14



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Research Project Periodic Log Entry Form

Project ID: R26-DS-010

Student ID: IT22294098

Name: Vidursha Prabagaran

Individual Component: Context-Aware Intelligent System for Automated ADL Extraction and Real-Time Documentation from Caregiver Voice Logs

Summary of RP work carried out during the reporting period

Date	RP Work
01.05.2026	Collected caregiver voice recording data using volunteer participants. Organised and structured the collected recordings according to the project data management plan, ensuring all audio files were correctly labelled and categorised by patient and recording session for use in subsequent pipeline stages.
02.05.2026	Conducted a second round of voice data collection using volunteers to expand the dataset. Subsequently set up the Google Colab environment and connected Google Drive for centralised dataset storage and model management. Created the project folder structure under HeyCareLog_Dataset with subfolders for audio, labels, models, results, and notebooks. Uploaded the master dataset CSV file to Google Drive for use across all pipeline notebooks.
03.05.2026	Worked on Notebook 01 - data_preprocessing.ipynb. Implemented the audio preprocessing pipeline to convert all raw recordings from M4A, MP3, OGG, and MP4 formats into standardized 16kHz mono WAV format using ffmpeg. Applied the noisereduce spectral subtraction library and achieved an average RMS energy reduction of 54.6% across all recordings. Generated waveform and spectrogram visualizations to verify noise removal. Implemented automatic SNR measurement using librosa to replace manually assigned noise labels, providing objective noise categorization. Completed additional preprocessing tasks and train, validation, and test data splitting, and saved all label files to Google Drive.



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04.05.2026	Worked on Notebook 02- 02_asr_comparison.ipynb. Conducted a systematic comparison of three Whisper model sizes on the held-out test recordings using Word Error Rate. Results showed WER of 37.68% for Whisper-tiny, 31.69% for Whisper-base, and 27.22% for Whisper-small. Whisper-small was selected as the Stage 2 ASR model. Also evaluated the impact of Stage 1 noise reduction on ASR accuracy, with noise reduction producing a 0.45% WER improvement on Whisper-small. Saved all results and comparison charts to Google Drive.
05.05.2026	Worked on Notebook 03 - disfluency_removal_V1.ipynb. Evaluated zero-shot performance of T5-small and BART-base. BART-base achieved BLEU of 0.3491 zero-shot, confirming its natural alignment with the text correction task. Fine-tuned BART-base on 968 caregiver training pairs, achieving BLEU of 0.8430 and ROUGE-L of 0.9339 at checkpoint-363. Fine-tuned T5-small for comparison, which achieved BLEU of 0.9872 but was later identified as overfitting.
06.05.2026	Conducted generalization testing on novel sentences. T5-small produced empty outputs for 4 out of 5 unseen sentences, confirming severe overfitting, and was therefore rejected. BART-base successfully generalized to 2 out of 5 sentences and was selected as the final model. Initiated Notebook 04 - disfluency_removal_V2.ipynb by applying three overfitting fixes: data augmentation to increase training pairs, reducing the learning rate from 5e-5 to 2e-5, and applying L2 weight decay of 0.01. Re-trained the BART-base v2 model.
07.05.2026	Completed BART v2 training. Checkpoint-684 was selected as the best model, achieving BLEU of 0.8525 and ROUGE-L of 0.9400. Generalisation testing showed 3 of 5 novel sentences correctly cleaned, an improvement over BART v1. Saved the model to the shared Google Drive folder and downloaded checkpoint-684 as a backup for frontend integration.
08.05.2026	Worked on Notebook 05 adl_extraction_V1.ipynb. Implemented and evaluated two ADL extraction approaches FLAN-T5-base using few-shot instruction following, and BERT-NER using domain-specific rule-based regex extraction covering 7 ADL fields. Evaluated both models on test rows using Precision, Recall, F1 Score, Alert Recall, and Alert F1 as evaluation metrics.



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09.05.2026	Completed ADL extraction evaluation. BERT-NER achieved Precision of 100%, F1 of 91.27%, and Alert Recall of 86.05%, significantly outperforming FLAN-T5-base which achieved F1 of 47.57% and Alert Recall of 0%. A combined model was also evaluated. BERT-NER was selected as the primary model based on superior metrics, zero RAM requirement, and full explainability. Saved all evaluation results to Google Drive.
10.05.2026	Developed the full pipeline demonstration notebook, 06_demo_pipeline.ipynb, integrating all completed stages into a single end-to-end workflow. Implemented an audio upload function to enable automatic processing through all pipeline stages with step-by-step output display. Tested the demo using three scenarios - normal day, medicine refusal, and fever detection - to verify correct pipeline behavior. Also initiated the development of the web application frontend to provide an interactive user interface for the pipeline demonstration.
11.05.2026	Attended a supervisor meeting and received feedback and suggestions on the research progress. Based on the supervisor's guidance, refined the frontend demo, successfully connected all pipeline models, and verified correct output generation across all stages. Prepared Progress Presentation 01 slides covering all completed pipeline stages and key evaluation results.
12.05.2026	Delivered Progress Presentation 01. Presented the individual research component covering all five completed pipeline stages and demonstrated the live pipeline demo showing end-to-end audio to ADL JSON output. PP1 successfully completed.
13.05.2026	Submitted the Master Sheet of Research as required. Completed and finalised the Data Analysis Report covering all data collection, preprocessing, modelling, and evaluation work carried out during the reporting period.

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14.05.2026	Initiated data collection planning for Progress Presentation 02. Requested additional data from Royal Care Nursing Home to obtain more ADL logs and expand the dataset, aiming to improve model performance, generalization, and reliability for the next phase of the research.

Supervisor's Comments

Student progressing well in research project: Yes / No
If No:

Project work to be carried out before next evaluation period:

Code base checked: Yes/No

Signature:

Date: 2026/05/14



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Research Project Periodic Log Entry Form

Project ID: R26-DS-010

Student ID: IT22289384

Name: Kaushalya Nagenthraraja

Individual Component: Personalized Sign Language Tutoring System for Caregivers

Summary of RP work carried out during the reporting period

Date	RP Work
01.05.2026	Analyzed the LSTM model training results and identified persistent overfitting issues despite regularization techniques. The model showed 61.58% test accuracy on the 18 WLASL-100 signs, but validation loss began diverging after 12 epochs. Recognized that the limited dataset size (approximately 400-500 videos for 18 signs) was insufficient for robust model generalization. Decided to expand the dataset by integrating the ASL Citizen dataset, which was previously evaluated but not utilized.
02.05.2026	Began downloading and preprocessing the ASL Citizen dataset to supplement the existing WLASL-100 data. Successfully obtained videos for 48 additional caregiver-relevant signs that were not available in WLASL-100 or had limited samples. The ASL Citizen dataset provided better video quality and more diverse signers (age, ethnicity, lighting conditions), which is crucial for building a robust caregiver-focused system. Combined both datasets to create a comprehensive vocabulary of 66 caregiver-specific signs.
03.05.2026	Conducted dataset integration and reorganization. Merged WLASL-100 and ASL Citizen videos into a unified folder structure with 66 sign classes containing a total of 2,696 video samples. Performed quality verification to ensure video readability and removed corrupted files. Implemented stratified train/validation/test splitting (70%/15%/15%) at the video level to prevent data leakage while maintaining balanced class distribution across all 66 signs.



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04.05.2026	Initiated MediaPipe landmark extraction pipeline for the complete merged dataset. Extracted hand landmarks (21 points per hand) and pose landmarks (33 points) from all 2,696 videos, representing each frame as a 258-dimensional feature vector (combining both hands and upper body pose). Encountered 12 unreadable videos in the test split, which were documented and excluded. The extraction process took approximately 6 hours and produced normalized landmark coordinated files for all successfully processed videos. Implemented advanced preprocessing techniques to enhance landmark data quality.
05.05.2026	Developed a stabilization algorithm to handle intermittent hand detection failures by interpolating missing landmarks across short gaps (< 5 frames). Applied coordinate normalization to make the features invariant to signer position, camera distance, and body size by centering landmarks relative to torso keypoints and scaling to a standard reference frame. This significantly improved consistency across different signers and recording conditions.
06.05.2026	Enhanced the feature representation by computing velocity and acceleration features from the normalized landmark sequences. For each landmark coordinate, calculated first-order (velocity) and second order (acceleration) temporal derivatives using finite differences. This expanded the feature dimensionality from 258 (position only) to 774 (position + velocity + acceleration), providing richer motion information critical for distinguishing dynamic sign language gestures. Verified the enhanced features maintained temporal continuity and showed clear motion patterns.
07.05.2026	Retrained the BiLSTM model on the expanded 66-class dataset using enhanced motion features (774 dimensions), class weighting, and stronger regularization (dropout = 0.4, L2 regularization). The model achieved 61.58% test accuracy and 60.49% macro F1-score, serving as a baseline for comparison. Although the model captured temporal patterns effectively, confusion between visually similar signs highlighted the need for alternative architectures. A GRU (Gated Recurrent Unit) model was then trained as a more parameter-efficient alternative. The GRU achieved 71.50% test accuracy and 70.72% macro F1-score across all 66 caregiver signs, showing a 10% improvement over the BiLSTM baseline. It also demonstrated better generalization with reduced overfitting, making it the strongest candidate for the final system.



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08.05.2026	Implemented a hybrid CNN+BiLSTM architecture to evaluate whether spatial feature extraction before temporal modeling could improve performance. The CNN layers were designed to capture local spatial patterns in landmark configurations, followed by BiLSTM for temporal sequence learning. The model achieved 67.18% test accuracy and 63.72% macro F1-score—better than baseline BiLSTM but underperforming compared to GRU. Observed increased training time and slight overfitting tendencies.
09.05.2026	Developed a Dynamic Time Warping (DTW) based feedback system for evaluating learner sign quality. Since neural networks provide only class predictions, DTW enables fine-grained comparison between a learner's gesture and reference demonstrations by measuring temporal alignment distance. Implemented specialized feature extraction for DTW comparison, focusing on hand motion trajectories and velocities rather than full body landmarks, ensuring speed-tolerant gesture evaluation.
10.05.2026	Calibrated the DTW scoring system using same-class and different-class gesture comparisons from the validation set. Established distance thresholds to convert raw DTW scores into interpretable feedback bands: Excellent (90-100), Good (70-89), Needs Improvement (50-69), and Poor (<50). Implemented corrective feedback generation that analyzes hand position errors, motion strength differences, and timing mismatches to provide actionable guidance such as "Left hand movement too weak" or "Hand trajectory deviates from reference."
11.05.2026	Conducted a comprehensive comparison of BiLSTM, GRU, and CNN+BiLSTM architectures using confusion matrices, learning curves, and per-class performance analysis. The GRU model achieved the best performance with 71.50% accuracy, making it the optimal choice for the final system. Model architectures, hyperparameters, and training procedures were documented to ensure reproducibility.Finalized the complete system implementation, including model export, DTW-based feedback functions, and MediaPipe task model integration. Exported the trained GRU model, label mappings, normalization parameters, and DTW calibration data for backend deployment. Also documented the preprocessing pipeline, feature engineering process, and evaluation results for future reference.



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12.05.2026	Compiled demonstration materials including sample predictions, confusion matrices, and feedback examples for the proposal presentation. Successfully attended and completed the proposal presentation.
13.05.2026	Following the proposal presentation, compiled and organized supervisor feedback and evaluation panel recommendations. Key suggestions included expanding real-world testing with sign language known person, improving feedback clarity for non-technical users, and exploring personalized learning adaptation mechanisms. Prepared the required data analysis report documenting the complete methodology from data acquisition to model evaluation. Additionally, compiled a master sheet of research papers covering five key areas: Sign Language Recognition Methods, Educational Learning Systems, Healthcare Applications, Behavioral Analysis in Elderly Care, and Computer Vision Techniques.
14.05.2026	Explored public dataset availability for elderly disease symptom recognition training. Systematically searched Kaggle, OpenNeuro, GitHub, and academic repositories for datasets related to Parkinson's disease, Alzheimer's/dementia, depression, and stroke in elderly populations. Evaluated each dataset based on modality (video/audio/EEG/MRI), size, accessibility, and suitability for training caregivers to recognize behavioral symptoms.

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Supervisor's Comments

Student progressing well in research project: Yes / No
If No:

Project work to be carried out before next evaluation period:

Code base checked: Yes / No

Signature:

Date: 2026/05/14



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Research Project Periodic Log Entry Form

Project ID: R26-DS-010

Student ID: IT22178640

Name: Dayana Priyadharshani Kumar

Individual Component: AI Assisted Caregiver Workforce Deterioration Detection and Adaptive Workload Redistribution

Summary of RP work carried out during the reporting period

Date	RP Work
16.05.2026	Reviewed PP1 feedback received after the presentation. Identified key areas flagged for improvement: individual behavioural pattern identification methodology and longitudinal data validation. Made a note of panel comments to address in PP2 planning. Organised existing project files and Google Drive folders in preparation for the next development phase.
19.05.2026	Read through the existing Data Analysis Report and PP1 checklist submissions to identify gaps and areas requiring further elaboration for PP2. Noted that the audio model fusion component and TILES-based longitudinal validation remain as pending work. Updated personal project timeline document with revised milestones for the two-month PP2 preparation window.
20.05.2026	Conducted a general review of literature related to occupational stress detection in healthcare workers. Re-read key sections from previously reviewed papers to refresh understanding of feature engineering approaches used in physiological signal processing. Noted references relevant to addressing the panel's comment on individual behavioural pattern identification.



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21.05.2026	Reviewed the research paper master sheet compiled for PP1. Cross-checked paper summaries against the panel feedback to identify which studies are most relevant to the behavioural pattern identification question. Added brief notes against three papers (Cho et al., 2023; Rana et al., 2024; Dogan & Akbulut, 2023) on how their methodologies relate to personalised baseline approaches
23.05.2026	Reviewed the existing Isolation Forest implementation notes from the Colab notebook. Re-examined the ablation study results (Nurse E4: FPR 1.00 → 0.00) to understand how to better present and justify the personalised baseline approach for PP2. Drafted a short explanation paragraph addressing the panel's question on individual behavioural pattern identification.
24.05.2026	Attended group meeting to discuss PP2 scope and responsibilities. Discussed integration requirements how individual components (stress detection, audio model, frontend) will be connected for the final system demonstration. Made notes on the expected backend integration tasks and agreed on a preliminary group task division.
25.05.2026	Reviewed MIMIC dataset access credentials and documentation to understand what data fields are available for potential use in PP2. Noted relevant tables related to nurse shift records and patient acuity metrics. This was exploratory — no data extraction performed. Documented observations for discussion with supervisor.
27.05.2026	Reviewed the CareSense frontend codebase to refresh familiarity with the component structure before resuming active development. Noted which components currently use hardcoded placeholder values versus real model outputs. Made a list of pages and data fields that will need updating once the PP2 backend integration is completed.

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29.05.2026	Read Mishra & Rizvi (2025) paper on LSTM-based speech emotion detection. Focused on the feature extraction methodology — MFCC coefficients and temporal windowing — to compare with the CNN-based approach used in the current audio stress classifier. Took notes on potential improvements to the audio model that could be explored in PP2.

Supervisor's Comments

Student progressing well in research project: Yes / No
If No:

Project work to be carried out before next evaluation period:

Code base checked: Yes / No

Signature:
Date: 2026-06-05



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Research Project Periodic Log Entry Form

Project ID: R26-DS-010

Student ID: IT22057488

Name: Harishalinee Elangovan

Individual Component: Edge-Enabled Skeletal Movement Analysis System for Real-Time Patient Observable Fall Risk Detection and Prediction

Summary of RP work carried out during the reporting period

Date	RP Work
15.05.2026	Reviewed the PP1 panel feedback and identified the key development tasks required for Phase 2: replacing the static JSON manifest with a live Flask API and adding a live camera input. Began reading Núñez-Marcos et al. (2017) on LSTM-based fall detection to assess whether a recurrent baseline would be worth implementing alongside ST-GCN for comparison purposes.
16.05.2026	Studied the BiLSTM architecture and its application to skeleton sequence classification. Set up Codes/models/lstm_baseline.py, defining the two-layer BiLSTM with hidden size 128 and dropout 0.3, using the same 14-joint 100-frame input format and subject-disjoint Cross-Subject split as the existing models
19.05.2026	Ran the BiLSTM training loop and observed convergence after approximately 22 epochs. Noted the model achieved lower high-risk recall compared to the ST-GCN, confirming that the flat joint vector input loses the anatomical graph structure. Saved the training log and confusion matrix to lstm_results/
23.05.2026	Resumed work after the second final examination. Finished the Transformer baseline implementation, ran training under the Cross-Subject protocol, and saved results to transformer_results/. Reviewed the outputs alongside the BiLSTM and ST-GCN results to confirm the ST-GCN still leads on high-risk recall.
24.05.2026	Resumed work after the second final examination. Finished the Transformer baseline implementation, ran training under the Cross-Subject protocol, and saved



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	results to transformer_results/. Reviewed the outputs alongside the BiLSTM and ST-GCN results to confirm the ST-GCN still leads on high-risk recall
25.05.2026	Started building the Flask API in Codes/api/app.py. Implemented the GET /api/manifest route and added CORS headers. Swapped the static file path in demoFeed.ts for the local Flask URL and confirmed the React dashboard loads patient data from the live endpoint
26.05.2026	Added the POST /api/predict route to the Flask API. Wired up the inference chain — MediaPipe extraction, 14-joint mapping, normalisation, and Fusion MLP prediction — and returned the result as a DemoEntry-shaped JSON response. Did a quick test with the laptop webcam to check the route responds without errors.
28.05.2026	Investigated latency in the inference loop. Found the normalisation step was the slowest part and rewrote it using vectorised NumPy instead of a per-frame loop. Also looked into camera options for the edge device, noting the Raspberry Pi Camera Module 3 and Intel RealSense D435 as candidates for future hardware integration.
29.05.2026	Began architectural planning for replacing the static JSON manifest with a live Flask HTTP API as outlined in the PP2 requirements. Outlined the two routes needed — GET /api/manifest and POST /api/predict — and reviewed how the one-line change in demoFeed.ts would switch the frontend to the live endpoint without modifying any UI components

Supervisor's Comments

Student progressing well in research project: Yes / No
If No:

Project work to be carried out before next evaluation period:

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Code base checked: Yes / No

Signature:

Date: 2026-06-05



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Research Project Periodic Log Entry Form

Project ID: R26-DS-010

Student ID: IT22294098

Name: Vidursha Prabagaran

Individual Component: Context-Aware Intelligent System for Automated ADL Extraction and Real-Time Documentation from Caregiver Voice Logs

Summary of RP work carried out during the reporting period

Date	RP Work
16.05.2026	Conducted a field visit to Birds of Paradise Elders Care Home to formally request permission for data collection in support of the research project. Presented the research objectives and data requirements to the facility management. Engaged in discussions with the care home staff regarding their current caregiving documentation practices and daily ADL recording procedures. The management reviewed the request positively and formally agreed to support the research by permitting data collection at their facility. This approval provides a valuable opportunity to expand the dataset with additional caregiver voice recordings from a second care home, which will contribute to improving model generalisation and reliability for PP2.
23.05.2026	Conducted a data collection session at Birds of Paradise Elders Care Home following the permission granted during the previous visit. Collected patient care data from existing handwritten records maintained by the care home staff. The records contained daily caregiver logs documenting ADL activities for elderly patients including hygiene, nutrition, medication, fluid intake, mood observations, and any reported symptoms. All collected information was carefully reviewed, manually transcribed, and organised according to the established dataset structure and naming conventions. The data was documented systematically to ensure accuracy and completeness. This handwritten record collection contributes valuable real-world caregiving data to support the dataset expansion required for the next phase of the research.



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24.05.2026	Organised and structured the handwritten caregiving records collected from Birds of Paradise Elders Care Home. Transcribed and categorised the data according to the established dataset labelling conventions, ensuring all ADL fields including hygiene, medication, fluid intake, mood, and symptoms were accurately captured. Updated the master dataset CSV file to incorporate the new entries and verified consistency with the existing dataset format.
25.05.2026	Reviewed the methodology section of the research project to assess alignment between the planned approach and the work completed to date. Identified areas requiring updates to reflect the expanded dataset from Birds of Paradise Elders Care Home. Cross-referenced the collected handwritten records against the ADL categories defined in the research framework to confirm coverage and completeness of the new data entries.
26.05.2026	Reviewed and studied two recently published research papers related to clinical NLP and ADL extraction to strengthen the literature foundation for the upcoming PP2 submission. Took detailed notes on relevant methodologies, evaluation approaches, and findings that could inform improvements to the existing pipeline stages. Identified gaps in the current literature that further support the novelty of the proposed research.
27.05.2026	Conducted a thorough review of all collected data from both Love and Elder Care and Birds of Paradise Elders Care Home. Assessed the overall dataset distribution across ADL categories, alert labels, and patient demographics to identify any imbalances or gaps. Documented observations and prepared a summary of the expanded dataset statistics to be included in the updated research report.
28.05.2026	Reviewed the overall project methodology and evaluated the progress of each pipeline stage against the PP2 objectives. Assessed the current state of the dataset, model training, and evaluation results. Updated personal research notes to reflect decisions made during the reporting period and prepared a structured task list for the remaining implementation work to be completed before PP2.



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Supervisor's Comments

Student progressing well in research project: Yes / No
If No:

Project work to be carried out before next evaluation period:

Code base checked: Yes / No

Signature:
Date: 2026-06-05



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Research Project Periodic Log Entry Form

Project ID: R26-DS-010

Student ID: IT22289384

Name: Kaushalya Nagenthraraja

Individual Component: Personalized Sign Language Tutoring System for Caregivers

Summary of RP work carried out during the reporting period

Date	RP Work
15.05.2026	Organized the findings from the elderly disease dataset search into a structured summary document. Categorized datasets by condition (Parkinson's, Alzheimer's/dementia, depression, stroke) and recorded details such as modality, sample size, number of subjects, accessibility, and licensing. Identified the most promising candidates for each condition and noted any limitations such as small sample sizes or restricted access.
16.05.2026	Went through the evaluation panel feedback from the proposal presentation in detail. Broke down the three main recommendations expanding real-world testing, improving feedback clarity for non-technical users, and exploring personalized learning adaptation into concrete sub-tasks. Created a prioritized action list with rough timelines to address each point systematically in the coming weeks.
19.05.2026	Focused on the misclassification patterns observed in the GRU model's confusion matrix. Identified sign pairs that were most frequently confused with each other, particularly signs with similar hand shapes or motion trajectories. Documented these problematic sign pairs and began thinking about whether additional training samples, augmentation, or architectural changes could help resolve the confusion in those specific classes.



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20.05.2026	Worked on understanding the nature of the misclassified sign pairs identified. Manually inspected a few video samples from the confused classes to visually verify whether the confusion was due to genuine visual similarity or landmark extraction noise. Found that several pairs shared nearly identical hand configurations and differed primarily in subtle wrist orientation or non-dominant hand placement, which the current feature set may not capture distinctly enough.
23.05.2026	Investigated data augmentation strategies that could improve the model's ability to distinguish visually similar signs. Looked into temporal augmentation techniques such as time stretching and frame dropping, as well as spatial augmentation like small random rotations and reflection of landmark coordinates. Considered which augmentations would be realistic for sign language data without distorting the linguistic meaning of the gestures.
25.05.2026	Implemented a basic temporal augmentation pipeline on the landmark sequences. Applied random frame dropping (removing up to 10% of frames) and time stretching (resampling sequences to slightly shorter or longer lengths) to generate augmented versions of training samples for the most underperforming sign classes. Verified that augmented sequences retained the general motion pattern of the original signs before incorporating them into the training pipeline.
26.05.2026	Retrained the GRU model incorporating the augmented samples for the identified low-performing sign classes. Observed a modest improvement in per-class accuracy for some of the previously confused pairs. Documented the before and after F1-scores for each affected class to quantify the impact of augmentation. Overall model accuracy showed marginal improvement, confirming that targeted augmentation helps but does not fully resolve the confusion for signs with near-identical spatial profiles.
27.05.2026	Shifted focus to the DTW-based feedback system. Re-examined the corrective feedback messages currently generated by the system and assessed how understandable they would be to a caregiver with no technical background. Found that phrases referencing trajectory deviation and landmark coordinates were too technical. Began rewriting the feedback templates using plain, action-oriented language that directly tells the user what to physically correct, such as describing hand placement relative to body parts rather than coordinate values.

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28.05.2026	Continued refining the feedback message templates. Expanded the set of corrective messages to cover a wider range of error types including hand height errors, dominant vs non-dominant hand confusion, and speed-related issues. Tested the revised messages against several DTW output scenarios to ensure the correct message is triggered for each error type. Made adjustments to the threshold conditions used to select messages to better reflect the actual error patterns observed in sample comparisons.
29.05.2026	Looked into personalized learning adaptation mechanisms as recommended by the evaluation panel. Surveyed approaches used in existing intelligent tutoring systems, including mastery-based progression, spaced repetition scheduling, and difficulty-adaptive content selection. Evaluated which of these approaches would be feasible to implement given the current system architecture and available learner performance data from the DTW scoring module.

Supervisor's Comments

Student progressing well in research project: Yes / No
If No:

Project work to be carried out before next evaluation period:

Code base checked: Yes / No

Signature:
Date: 2026-06-05



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Research Project Periodic Log Entry Form

Project ID: R26-DS-010

Student ID: IT22178640

Name: Dayana Priyadharshani Kumar

Individual Component: AI Assisted Caregiver Workforce Deterioration Detection and Adaptive Workload Redistribution

Summary of RP work carried out during the reporting period

Date	RP Work
01.07.2026	Reviewed the overall research problem and the objectives defined for the project. Re-examined the relationship between workload, individual performance, physiological indicators, and contextual factors within the intended research setting. Reviewed the previously proposed methodology to identify areas that required stronger empirical support. Identified the need for a dataset containing sufficiently rich behavioural, physiological, contextual, or performance-related information. Documented the main research questions that needed to be addressed before finalizing the modelling methodology.
03.07.2026	Reviewed the previous research approach associated with Hossien's work in relation to the current research objectives. Examined the variables, methodology, modelling assumptions, and evaluation strategy used in the previous work. Considered which aspects of the approach could potentially support the current research direction. This review helped establish a more critical approach to using previous research rather than directly reproducing an existing methodology.
04.07.2026	Continued reviewing relevant literature concerning workload, performance assessment, physiological sensing, behavioural indicators, and machine-learning-based analysis. Compared the general methodologies used by previous studies with the requirements of the current research problem. Examined how multimodal data had been used to identify patterns associated with workload and performance.



	Identified limitations in approaches that relied on a single data modality or narrowly defined experimental settings. Considered the importance of combining appropriate data sources while maintaining interpretability and avoiding unnecessary complexity.
05.07.2026	Defined preliminary criteria for evaluating whether a dataset was suitable for the research. These criteria were later used to assess the suitability of TILES-2018 for the proposed research direction. Considered the importance of participant-level information, temporal information, and repeated observations for constructing a reliable experimental design. Investigated whether the intended research would require multimodal information rather than relying on a single source
07.07.2026	Reviewed the available TILES-2018 documentation and dataset descriptions to investigate whether the information required by the existing Hossien-based model was available. Compared the expected input variables of the existing model with the types of information reported as available in TILES-2018. Examined potential differences in data structure, participant information, sensing modalities, contextual information, and outcome definitions. Documented the variables that appeared potentially transferable and those requiring further verification after direct dataset access.
08.07.2026	Critically evaluated the original plan of transferring the Hossien-trained model to TILES-2018 through fine-tuning. Reviewed the requirements that must be satisfied for meaningful fine-tuning, including compatible input features, feature representations, target definitions, and sufficiently related data distributions. Identified potential problems that could arise if the TILES data did not contain equivalent variables or measurement structures. Based on the compatibility assessment, developed an alternative methodological direction in which TILES-2018 would be investigated as an independent research dataset rather than being treated simply as a fine-tuning source. Considered how the research objectives could be reformulated around the information actually available within TILES-2018.
09.07.2026	Modified the existing backend structure to separate the model-inference logic from the rest of the application services. Refactored the model-related code so that the preprocessing and prediction components could be replaced without affecting the main application workflow. Updated the backend input structure to make the model features configurable rather than permanently tied to the Hossien dataset feature set. Separated the existing Hossien model artefacts from the new model-development workflow to prevent the two experimental pipelines from becoming mixed. Added or updated configuration points for model paths, feature definitions, and prediction outputs.



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10.07.2026	Modified the existing frontend model-output components to make the displayed prediction information independent of the previous Hossien model. Updated the frontend data-handling logic so that prediction results could be received from a revised backend response structure. Adjusted the prediction display components to accommodate changes in model outputs without redesigning the complete interface. Updated the application data flow so that future TILES-derived inputs could be passed through the backend inference pipeline.
12.07.2026	Prepared the experiment-tracking structure for recording dataset versions, feature configurations, model configurations, evaluation metrics, and ablation experiments. Added the initial preprocessing and modelling pipeline structure without fixing the final features or target before examining the TILES data. Prepared separate locations for training data, validation data, testing data, and future inference/demo data. Updated the backend configuration to allow a finalized model artefact to be connected after the modelling experiments were completed.
13.07.2026	Reorganized the research methodology section to reflect the transition from the initial Hossien-based approach toward the proposed TILES-2018 investigation. Documented the original data-processing, feature-selection, modelling, and ablation workflow used during the first phase of the research. Added the methodological justification for investigating TILES-2018 as a new dataset rather than assuming that the previously developed model could be directly transferred.
15.07.2026	Revised the research-gap discussion to clearly distinguish the completed first-phase work from the new research direction being developed using TILES-2018. Documented the limitation of assuming that a model developed from one dataset could automatically be fine-tuned on a structurally different dataset. Added the planned dataset-compatibility investigation as a methodological decision within the research framework.



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Supervisor's Comments

Student progressing well in research project: Yes / No
If No:

Project work to be carried out before next evaluation period:

Code base checked: Yes/No

Signature: 

Date: 18.07.2026



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26 REGULAR RP

Research Project Periodic Log Entry Form

Project ID: R26-DS-010

Student ID: IT22057488

Name: Harishalinee Elangovan

Individual Component: Edge-Enabled Skeletal Movement Analysis System for Real-Time Patient Observable Fall Risk Detection and Prediction

Summary of RP work carried out during the reporting period

Date	RP Work
01.07.2026	Began studying the NTU RGB+D 120 dataset in detail, reviewing the skeleton sequence format and subject metadata. Identified the subset of action classes relevant to fall detection: falling forward, falling backward, falling sideways, and chest pain. Noted the 25-joint NTU skeleton structure and planned the mapping strategy to reduce it to 14 joints aligned with the MediaPipe Pose output for consistent model input across both datasets.
02.07.2026	Started drafting the related work section of the research paper. Reviewed five existing fall detection papers to identify research gaps: Núñez-Marcos et al. (2017) on LSTM-based fall detection, Adhikari et al. (2019) on wearable sensor approaches, and Kwolek and Kepski (2014) on depth camera methods. Noted that most existing systems rely on dedicated depth sensors or wearables rather than a standard RGB webcam, identifying this as the key gap the proposed system addresses.
03.07.2026	Read Yan et al. (2018) "Spatial Temporal Graph Convolutional Networks for Skeleton-Based Action Recognition" in full. Studied the graph adjacency matrix construction, the spatial graph convolution formulation, and the temporal convolution stack. Noted the partitioning strategies and decided to adopt spatial configuration partitioning for the ST-GCN implementation as it gave the strongest reported results on NTU RGB+D. Added this paper as a primary citation in the research paper methodology section.



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05.07.2026	Reviewed three papers on transfer learning for skeleton-based recognition to understand how a model pre-trained on a large, labelled dataset can be adapted to a smaller in-house dataset via fine-tuning. Noted that freezing the backbone graph convolutional layers for the first few epochs and training only the classification head prevents catastrophic forgetting. Documented this as the planned fine-tuning strategy for Model C and added supporting references to the research paper.
07.07.2026	Studied the UR Fall Detection dataset structure. Identified the fall and ADL sequences relevant to the binary classification task and noted the subject count and sequence lengths. Planned the subject-disjoint Cross-Subject split to ensure no subject appears in both training and validation sets, preventing data leakage and producing an honest generalization estimate. Documented the dataset description and split rationale in the research paper dataset section.
08.07.2026	Began researching suitable camera hardware for the edge deployment scenario. Compared three options: standard USB webcam (Logitech C920), Raspberry Pi Camera Module 3, and Intel RealSense D435 depth camera. Evaluated each against four criteria: cost, edge device compatibility, MediaPipe Pose support, and ward environment suitability. Documented the comparison in the research paper hardware section and identified the standard USB webcam as the most practical choice for cost-effective edge deployment without depth sensor dependency.
09.07.2026	Reviewed the MediaPipe Pose landmarker documentation and identified the 33-landmark output format. Mapped 14 anatomically meaningful joints from the full 33-landmark set for compatibility with the ST-GCN graph topology. Verified that MediaPipe provides per-landmark visibility scores usable for flagging low-confidence frames during preprocessing. Sketched the joint adjacency graph to confirm all bone connections are anatomically valid and included the joint diagram in the research paper system architecture section.
10.07.2026	Drafted the system architecture section of the research paper. Described the four-stage pipeline: (1) RGB video capture via USB webcam, (2) real-time skeleton extraction using MediaPipe Pose, (3) ST-GCN inference on the 90-frame sliding window, and (4) late fusion with kinematic features to produce a fall risk score. Created a pipeline diagram illustrating data flow from camera input to risk output and included it in the paper draft.



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11.07.2026	Defined the target input shape as (90 frames × 14 joints × 4 channels: x, y, z, visibility) representing a 3-second sliding window at 30 fps. Outlined the normalisation strategy: centring the skeleton at the mid-hip joint and scaling by torso height to make representations subject-invariant across different body sizes. Wrote pseudocode for the temporal resampling function to handle variable-length clips via linear interpolation and included it in the paper methodology section.
12.07.2026	Investigated edge deployment constraints for a laptop CPU without a dedicated GPU. Reviewed PyTorch CPU inference benchmarking practices and identified the preprocessing pipeline and ST-GCN forward pass as the two dominant latency contributors. Planned to benchmark each stage separately using time.perf_counter() over 100 runs with a 10-run warmup. Drafted the evaluation metrics section of the research paper covering accuracy, macro F1, ROC-AUC, and latency per window as the four primary evaluation criteria.
14.07.2026	Set up the backend project directory structure under backend/src/. Created placeholder files for data_processor.py, features.py, models/stgcn.py, and models/fusion.py. Installed core dependencies into a virtual environment: PyTorch 2.x, MediaPipe 0.10, OpenCV, scikit-learn, and NumPy. Confirmed the environment runs correctly on CPU fallback to match the target edge deployment hardware. Tested a basic OpenCV camera feed on the development laptop to verify frame capture at 30fps.
15.07.2026	Designed the 18 kinematic feature set for the late fusion branch. Features include body sway speed, trunk tilt angle, hip velocity, shoulder velocity, ankle velocity, centre-of-mass horizontal and vertical shift, step symmetry index, and per-joint angular velocity for key limb pairs. Reviewed supporting literature confirming these features capture fall-precursor motion patterns not encoded in raw joint coordinates. Documented all 18 features with formulas in the research paper and as function stubs in src/features.py.

Supervisor's Comments

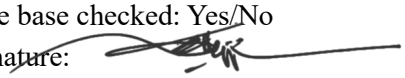
Student progressing well in research project: Yes / No

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If No:

Project work to be carried out before next evaluation period:

Code base checked: Yes/No
Signature: 
Date: 18.07.2026



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Research Project Periodic Log Entry Form

Project ID: R26-DS-010

Student ID: IT22294098

Name: Vidursha Prabagaran

Individual Component: Context-Aware Intelligent System for Automated ADL Extraction and Real-Time Documentation from Caregiver Voice Logs

Summary of RP work carried out during the reporting period

Date	RP Work
01.07.2026	Conducted a follow-up visit to Love and Elder Care to formally request permission for an expanded round of data collection, covering both the male and female caregiving wards. Presented the updated data requirements to facility management, outlining the need for additional handwritten ADL records and caregiver voice recordings to support the accent- and disfluency-adaptation stages of the pipeline. The request was approved, enabling a four-day collection schedule across both wards.
02.07.2026	Conducted the first data collection session at the male caregiving ward of Love and Elder Care. Gathered handwritten ADL records maintained by caregiving staff and recorded caregiver voice logs describing daily patient care activities using a portable voice recorder. Data was labelled by ward and date to maintain traceability within the dataset structure.
03.07.2026	Conducted a second data collection session at the male caregiving ward, continuing to gather handwritten caregiving records and caregiver voice recordings. Cross-checked the newly collected entries against the previous day's records to avoid duplication and ensure consistent coverage of ADL categories across shifts.
04.07.2026	Compiled the narrated audio recordings and their corresponding transcribed text into a structured Excel sheet, formatting the entries in line with the established dataset conventions used in earlier stages of the project.



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07.07.2026	Conducted an initial fine-tuning trial using the compiled caregiving audio-text dataset to assess whether it could support the ASR accent-adaptation stage directly, given its closer match to real caregiving speech patterns.
09.07.2026	Evaluated the fine-tuning results from the caregiving dataset and found the resulting word error rate to be substantially high, indicating that the dataset size was insufficient to reliably fine-tune the ASR model. Documented this as a data-scale limitation rather than a methodology failure.
10.07.2026	In response to the identified data-scale limitation, evaluated available public alternatives and confirmed Svarah (AI4Bharat, 6,656 transcribed utterances from 117 Indian English speakers) as the most appropriate accent-adaptation dataset, given the continued absence of a Sri Lankan English speech corpus of sufficient scale. Formally adopted Svarah as the primary training resource for the ASR stage going forward, while retaining the caregiving recordings as a small real-world validation reference for later stages.
11.07.2026	Received a letter from Remark Elder Care acknowledging the earlier research visit and discussion held with the facility, in which CEO Mr. Vasantha requested the research team to write a blog or article detailing the research approach undertaken in the industry and the documentation gaps identified during the study. Mr. Vasantha expressed interest in publishing the article on the Remark Elder Care website under the authors' names, offering it as a formal reference for the research project. Separately, reviewed a research paper examining automatic speech recognition performance on Indian-accented English, which reported consistent word error rate degradation across multiple ASR models on Indian English speech, taking detailed notes to support the justification for the Svarah-based accent-adaptation approach.
12.07.2026	Reviewed a second research paper analysing word error rate patterns in Indian English speech recognition, comparing its evaluation methodology against the first paper to strengthen the literature basis for the accent-adaptation stage.

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14.07.2026	Reassessed the framing of the research novelty in light of the literature reviewed. Identified that reporting a single fine-tuning result is consistent with prior published work, whereas systematically varying one training setting at a time and observing the resulting trend across multiple fine-tuning runs had not been demonstrated for this accent group. Formally documented this multi-run ablation design as the core novelty contribution for the ASR stage, directly addressing feedback expected at PP2.
15.07.2026	Began implementation of the data-fraction ablation experiment on the Svarah dataset, configuring the LoRA fine-tuning pipeline to train separately on 10%, 25%, 50%, and 100% of the training data using corrected shuffled, seeded sampling. Initiated the first set of training runs under this protocol.

Supervisor's Comments

Student progressing well in research project: Yes / No
If No:

Project work to be carried out before next evaluation period:

Code base checked: Yes/No
Signature:
Date: 18.07.2026



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26 REGULAR RP

Research Project Periodic Log Entry Form

Project ID: R26-DS-010

Student ID: IT22289384

Name: Kaushalya Nagenthraraja

Individual Component: Personalized Sign Language Tutoring System for Caregivers

Summary of RP work carried out during the reporting period

Date	RP Work
01.07.2026	Conducted a detailed review of the GRU model developed for PP1. It showed that the v1 stacked GRU models contained approximately 200,000–330,000 parameters, while the training dataset consisted of only 1,654 sequences across 59 classes. This indicated that the models were relatively large compared to the available training data, increasing the risk of overfitting and limiting their ability to generalize to unseen samples.
02.07.2026	I reviewed relevant literature on lightweight sequence-learning models suitable for small time-series datasets. Based on the review, a Temporal Convolutional Network (TCN) with approximately 45K parameters and a compact SPOTER-style Transformer with approximately 83K parameters were selected for comparative evaluation.
04.07.2026	I then added a params_per_train_sample check to the training script to monitor the relationship between model size and the number of available training samples. A threshold of approximately 50 parameters per training sample was introduced to identify models that may be unnecessarily large for the available dataset.



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05.07.2026	Evaluated different spatial normalization approaches and identified an issue affecting the recognition of location-dependent signs. The previous wrist-based recentering removed important spatial information between the hands and the upper body. This information was found to be relevant for distinguishing signs such as "pain", "sick", and "tired", where the signing location contributes to the meaning.
06.07.2026	I had to modify the spatial normalization process by using the midpoint between the shoulders as the reference point. The complete upper-body skeleton was then rotated using a common transformation to maintain the relative positions between the body and hands, replacing the previous separate rotation applied to individual body parts.
07.07.2026	Implemented scale normalization using shoulder width as the reference measurement instead of a hand-based measurement. This was selected to provide a more consistent representation of upper-body movements, particularly for signs performed in a seated caregiver context.
08.07.2026	Conducted a visual analysis of motion patterns in the temporal sequences to identify unusually long or irregular samples. A significant anomaly was identified in a "yes" sign sequence containing 664 frames, compared with a median sequence length of approximately 71 frames. The sequence also contained several separate high-motion regions, suggesting that multiple repetitions had been combined into a single file.
09.07.2026	I had to then apply a dataset-wide filtering criterion to identify sequences exceeding three times the median sequence length, corresponding to more than 213 frames. The process identified 11 sequences requiring further inspection and possible segmentation and also identified an issue related to the file naming structure of the ASL Citizen dataset. The dataset uses similar filenames across different class folders, meaning that the same filename may refer to different valid samples depending on its class.
11.07.2026	We visited the Remark Elder Care Home to discuss the progress of our research and obtain an official letter validating our previous onsite visits to the care facility. During the discussion, the CEO, Mr. Vasantha, also requested that we prepare a blog/article describing our research in the caregiving sector, including the key gaps and challenges we identified throughout the research process. He expressed interest in publishing the article on the care home's website under our names and indicated that the published article could also serve as a reference for our academic work.

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12.07.2026	Modified the segmentation and data-removal process to identify samples using both the class name and filename. This ensured that only the intended samples were modified or removed and prevented valid samples with matching filenames in other classes from being affected and executed the corrected segmentation process on the identified sequences. Ten of the 11 flagged videos were successfully segmented, producing 25 clean replacement sequences. This resulted in a net increase of 15 valid training sequences in the dataset
14.07.2026	Finalized the Layer 1 preprocessing pipeline and verified the resulting dataset. The processed dataset contained 59 classes and 2,364 sequences, divided into training, validation, and testing sets using a stratified 70/15/15 split. The final feature representation contained 60 temporal frames with 147 features per frame, resulting in a feature shape of (60, 147).

Supervisor's Comments

Student progressing well in research project: Yes / No

If No:

Project work to be carried out before next evaluation period:

Code base checked: Yes/No

Signature: 

Date: 18.07.2026



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26 REGULAR RP

Research Project Periodic Log Entry Form

Project ID: R26-DS-010

Student ID: IT22178640

Name: Dayana Priyadharshani Kumar

Individual Component: AI Assisted Caregiver Workforce Deterioration Detection and Adaptive Workload Redistribution

Summary of RP work carried out during the reporting period

Date	RP Work
17.07.2026	Received the permission for accessing the TILES-2018 dataset via FileZilla through a dedicated server. Received the credentials for accessing and did a structured examination of the TILES-2018 dataset and its available data sources. Reviewed the relevant README documentation in each sub folders to determine whether a suitable research target could be defined. Investigated the Fitbit and Omsignal-related data sources to determine their usability for physiological feature extraction. Examined the relationship between Jelly and Owls/location information to assess whether relational or proximity-based features could be constructed.
18.07.2026	Defined the candidate prediction target based on the actual information available in the TILES-2018 data rather than assuming that a particular target was present. Evaluated the suitability, completeness, and research relevance of the available target information. Finalized the candidate feature groups that could reasonably be used for modelling. Documented the reasons for including relevant physiological, workload, contextual, or relational information where supported by the dataset. Identified variables that could introduce leakage or were unsuitable for model development. Established a participant-level data-splitting strategy to separate training, validation, testing, and application-demonstration data.
19.07.2026	Developed the preprocessing workflow required to transform the selected TILES-2018 sources into a modelling-ready dataset. Addressed the required data-cleaning, missing-value, temporal aggregation, and feature-transformation procedures.



	Ensured that preprocessing operations capable of learning from the data were fitted using the training portion only. Developed the initial physiological and workload-related feature representation for the primary modelling approach.
20.07.2026	Established a baseline machine-learning experiment using the prepared TILES-2018 modelling dataset. Trained an initial interpretable model to provide a reference point for subsequent model comparisons. Developed the primary feature-based model using the selected physiological and workload information. Developed the second model configuration incorporating the additional relational feature group where the data supported its construction. Compared the model configurations using the predefined evaluation methodology. Performed the initial ablation comparison to investigate whether the additional feature group contributed meaningful predictive information.
22.07.2026	Reviewed the evaluation results to ensure that the comparison was based on the same data-splitting and preprocessing procedures. Documented the experimental methodology, including dataset selection, target definition, feature construction, preprocessing, and evaluation procedures. Began organizing the results into tables and figures suitable for the research paper. Interpreted the model comparison without relying solely on a single performance metric. Identified methodological limitations and areas requiring additional verification. Used the documented results to establish a clear connection between the experimental design and the research objectives.
24.07.2026	Investigated the behaviour of the stronger candidate model using an appropriate explainability approach. Examined the contribution of important features to the model's predictions. Investigated representative prediction errors to understand cases where the model performed poorly. Reviewed false-positive and false-negative cases where applicable to the selected prediction task. Considered whether participant or contextual differences could explain difficult prediction cases. Documented observations that could contribute to the interpretation and limitations sections of the research.
25.07.2026	Compared the candidate models using performance, interpretability, robustness, and deployment considerations. Selected the model that provided the most appropriate balance between predictive performance and the requirements of the research problem. Saved and versioned the selected model as the candidate deployment model. Documented the exact input variables and feature transformations required by the selected model.



	Defined the expected input data types and preprocessing sequence required during inference.
26.07.2026	Reviewed the experimental findings in relation to the original research objectives. Identified the main implications of the modelling and ablation results. Examined limitations associated with the TILES-2018 dataset, selected modalities, target definition, and modelling approach. Considered the limitations of using historical TILES data to simulate application inputs rather than receiving live wearable measurements. Documented the distinction between research experimentation and application demonstration.
27.07.2026	Used the finalized model-input specification to determine which information needed to be represented in the application database. Identified the fields required for model inputs, predictions, timestamps, model metadata, and historical results. Distinguished between raw or source information, processed model features, prediction records, and application-level user information. Designed the database structure around the actual requirements of the selected model rather than assumptions made before model selection. Considered how historical predictions could be retained to support trend visualization.
28.07.2026	Integrated the selected model into the backend inference workflow. Implemented the processing required to transform application inputs into the format expected by the frozen model. Connected the model to an inference endpoint capable of receiving the defined input structure. Configured the endpoint to return the model prediction and relevant output information. Verified that inference used the same preprocessing logic established during model development.

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Supervisor's Comments

Student progressing well in research project: Yes / No
If No:

Project work to be carried out before next evaluation period:

Code base checked: Yes/No

Signature:

Date: 03.08.2026



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26 REGULAR RP

Research Project Periodic Log Entry Form

Project ID: R26-DS-010

Student ID: IT22057488

Name: Harishalinee Elangovan

Individual Component: Edge-Enabled Skeletal Movement Analysis System for Real-Time Patient Observable Fall Risk Detection and Prediction

Summary of RP work carried out during the reporting period

Date	RP Work
16.07.2026	Began implementing data_processor.py. Coded the joint mapping function to reduce the 25-joint NTU skeleton to 14 joints and the 33-landmark MediaPipe output to the same 14 joints. Implemented the mid-hip centering normalization and torso-height scaling. Verified the output shape (90, 14, 4) matches the expected ST-GCN input format by running unit checks on sample NTU skeleton files.
17.07.2026	Implemented the ST-GCN graph topology in models/stgcn.py. Defined the 14-joint adjacency matrix encoding 13 bone connections and implemented the spatial graph convolution layer with learnable edge weights. Confirmed the graph structure matches the anatomical joint connectivity by visualizing the adjacency matrix as a heatmap and cross-checking against the MediaPipe skeleton diagram.
18.07.2026	Implemented the 18 kinematic feature extraction functions in features.py. Completed body sway speed, trunk tilt angle, hip and shoulder velocity calculations using frame-difference vectors. Tested each function on a synthetic skeleton sequence and confirmed output dimensions match the expected (18,) feature vector. Integrated the feature extractor into the data loading pipeline.
19.07.2026	Implemented the LateFusionNetwork in models/fusion.py. The network takes the 128-dimensional ST-GCN embedding concatenated with the 18 kinematic features as input and passes it through a three-layer MLP with batch normalization and dropout to produce a two-class risk output. Verified the forward pass runs without errors on random input tensors and confirmed output logits are of the correct shape (batch, 2).



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21.07.2026	Investigated USB webcam options suitable for edge deployment in a ward environment. Evaluated the Logitech C920 for resolution, frame rate, and low-light performance. Tested the camera connection on the development laptop using OpenCV VideoCapture and confirmed stable 30fps capture at 1280×720 resolution. Noted that MediaPipe Pose runs reliably at this resolution without significant frame drops on CPU.
22.07.2026	Set up the camera recording script record_clip.py for structured in-house data collection. Implemented keyboard controls (R to start recording, S to stop and save, Q to quit) with live MediaPipe skeleton overlay on the preview window. Configured automatic file naming by subject ID and label (fall/normal) under Datasets/Webcam/raw_videos/. Conducted a test recording session to verify the script saves clips correctly with proper skeleton detection.
23.07.2026	Implemented the model training loop in src/train.py. Set up the subject-disjoint Cross-Subject split, class-weighted sampler to handle fall/normal class imbalance, Adam optimizer with weight decay 1e-4, and early stopping with patience 10. Added training augmentation including random joint noise injection and temporal jitter. Ran a smoke test on a small NTU data subset to confirm the loop runs without errors and loss decreases over the first few epochs.
24.07.2026	Evaluated camera placement angle and distance for optimal full-body skeleton detection. Tested three positions at 1.5m, 2m, and 2.5m from the subject at approximately 1.2m height. Confirmed 2m distance at mid-torso camera height captures the full body from head to feet consistently. Documented the final camera placement specifications to be used for all subject recording sessions.
25.07.2026	Began full Model A training on the combined NTU RGB+D and UR Fall Detection dataset under the subject-disjoint Cross-Subject protocol. Monitored training loss and validation macro F1 per epoch. Observed steady convergence across early epochs. Saved model checkpoints at each epoch where validation F1 improved. Training is ongoing — final evaluation metrics to be recorded upon completion.
26.07.2026	Conducted the first in-house recording session with Subject 1 (family member). Recorded 15 fall clips covering side falls, forward falls, and backward falls, and 20 normal activity clips covering walking, sitting, standing, and bending. Verified



	each clip using the live MediaPipe skeleton overlay to confirm all 14 joints are detected consistently throughout each sequence.
28.07.2026	Implemented extract_webcam_mediapipe.py to process recorded video clips. The script runs MediaPipe Pose frame-by-frame on each clip, maps the 33 landmarks to 14 joints, and saves the skeleton sequence as a compressed .npz file. Processed all Subject 1 clips and confirmed output shape (T, 14, 4) for each sequence. Reviewed per-landmark visibility scores to identify any low-confidence frames requiring re-recording
29.07.2026	Continued monitoring Model A training progress. Validation macro F1 showing consistent improvement across epochs indicating the model is learning discriminative fall versus normal patterns from the NTU and UR datasets. Reviewed intermediate confusion matrices and noted high recall on fall class is being achieved progressively. Full training expected to complete within the next few days.
30.07.2026	Implemented parse_webcam.py to aggregate all per-clip .npz skeleton files into unified training arrays. Applied clip normalization, enforced the 90-frame temporal window via linear interpolation, and extracted the 18 kinematic features for each sequence. Saved merged arrays webcam_data.npy, webcam_features.npy, webcam_labels.npy, and webcam_subjects.npy to the processed data directory for use in Model C fine-tuning.
31.07.2026	Validated the Subject 1 webcam dataset by analysing hip Y-coordinate trajectories across all clips. Fall clips show a clear downward displacement pattern while normal clips remain stable, confirming the recordings capture fall events correctly at the skeleton level. Began planning the fine-tuning pipeline (finetune_webcam.py) for Model C, outlining the freeze-then-unfreeze training strategy where backbone layers are frozen for the first 3 epochs to preserve pre-trained representations before full fine-tuning begins.

Supervisor's Comments

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Student progressing well in research project: Yes / No
If No:

Project work to be carried out before next evaluation period:

Code base checked: Yes/No
Signature:
Date: 03.08.2026



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26 REGULAR RP

Research Project Periodic Log Entry Form

Project ID: R26-DS-010

Student ID: IT22294098

Name: Vidursha Prabagaran

Individual Component: Context-Aware Intelligent System for Automated ADL Extraction and Real-Time Documentation from Caregiver Voice Logs

Summary of RP work carried out during the reporting period

Date	RP Work
16.07.2026	Began implementation of the data-fraction ablation experiment on the Svarah dataset, configuring the LoRA fine-tuning pipeline to train separately on 10%, 25%, 50%, and 100% of the training data using corrected shuffled, seeded sampling. Initiated the first set of training runs under this protocol.
17.07.2026	Continued running the data-fraction ablation experiment across the remaining data splits, monitoring training stability and recording the word error rate achieved at each fraction for comparative analysis.
18.07.2026	Completed the data-fraction ablation experiment. Compiled the resulting WER values into a comparison table and chart, and drafted an analysis paragraph describing the trend of improving WER with increasing data volume, with gains beginning to plateau at higher data fractions , providing an early indicator of how much real caregiving data will likely be needed at Stage 4.
19.07.2026	Began the second ablation experiment, varying LoRA rank (4, 8, 16, 32) at fixed 100% data usage, to examine the trade-off between adaptation capacity and transcription accuracy for a lightweight, deployment-realistic model.
22.07.2026	Continued running the LoRA rank ablation experiment across the remaining rank values, recording WER at each configuration for comparison against the data-fraction ablation results.



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23.07.2026	Completed the LoRA rank ablation experiment. Compiled the results into a comparison table and chart, and drafted an analysis paragraph assessing whether increasing rank continued to improve accuracy or plateaued beyond a certain adaptation capacity - completing both planned ablation studies for the ASR stage.
24.07.2026	Cross-referenced the ablation results against the small caregiving audio-text dataset collected earlier in the month, using it as a real-world validation reference to assess how the Svarah-adapted model is expected to generalise to genuine caregiving speech ahead of full Stage 4 data collection.
25.07.2026	Requested and obtained access to the DisfluencySpeech dataset to support development of the auto-editing (disfluency-removal) stage of the pipeline. Reviewed its structure to plan integration with hand-constructed domain-specific examples for healthcare-related phrasing.
26.07.2026	Requested and obtained access to the DISCO dataset as a second resource for the auto-editing stage. Reviewed its annotation conventions against DisfluencySpeech to determine how the two datasets could be combined for training.
27.07.2026	Cross-referenced the DisfluencySpeech and DISCO datasets against caregiving and ADL-related vocabulary needs, identifying gaps where general conversational disfluency examples did not adequately represent healthcare-specific phrasing, and planned the preprocessing steps required before Stage 3 fine-tuning.
28.07.2026	Formally documented the ablation-based novelty framing in the research methodology section, describing how the data-fraction and LoRA rank ablation experiments distinguish this work from prior studies that report only a single fine-tuning result for Indian-accented ASR - strengthening the argument for PP2.
29.07.2026	Updated the master dataset documentation to consolidate the Svarah-based ASR ablation results with the small caregiving validation dataset, verifying consistency across formats, labelling conventions, and file structure ahead of PP2 review.

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30.07.2026	Reviewed overall progress against the PP2 objectives, evaluating the completed ASR ablation experiments, the justified pivot from limited caregiving data to Svarah, and the acquisition of DisfluencySpeech and DISCO for the upcoming auto-editing stage. Assessed the project as on track with a clearly evidenced methodology.

Supervisor's Comments

Student progressing well in research project: Yes / No
If No:

Project work to be carried out before next evaluation period:

Code base checked: Yes/No
Signature:
Date: 03.08.2026



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26 REGULAR RP

Research Project Periodic Log Entry Form

Project ID: R26-DS-010

Student ID: IT22289384

Name: Kaushalya Nagenthraraja

Individual Component: Personalized Sign Language Tutoring System for Caregivers

Summary of RP work carried out during the reporting period

Date	RP Work
16.07.2026	I began the Layer 2 Phase 1 development by focusing on data augmentation to improve the robustness of the recognition model. I implemented whole-sequence rotation jitter (±8°), scale jitter (±5%), and temporal-speed variation (±15%) through double resampling. These augmentation techniques were applied only to the training data to avoid affecting the validation and test results.
18.07.2026	I also extended the feature representation by calculating velocity and acceleration from the landmark sequences using first- and second-order temporal derivatives. These kinematic features were combined with the original positional features, increasing the representation to 441 features per frame.
19.07.2026	Addressed the class imbalance observed in the dataset by implementing a customized Focal Loss function. Class-specific weighting was incorporated into the loss calculation, together with focal modulation ($\gamma = 2.0$) and 0.1 label smoothing, to improve the model's ability to learn from classes with fewer or more difficult samples.
20.07.2026	I also had to make improvements to the Colab training environment to reduce the risk of losing training progress when sessions were interrupted. A MacroF1EarlyStoppingResumable mechanism was implemented to save the best-performing model state to Google Drive whenever an improvement in validation performance was detected.



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21.07.2026	I then had to identify and correct an issue where I had coded the same content twice in the notebook causing duplication which affected the results. The ensemble evaluation code had been repeated multiple times, which resulted in duplicated outputs in the evaluation tables. I had to resolve the errors and the order of the code and verified that the evaluation sections were executing correctly.
22.07.2026	Initiated the baseline training experiments for the Layer 2 TCN model. During the initial experiments, I observed considerable differences in the evaluation results between separate training runs, even though the same code and dataset were used. This raised a concern about reproducibility that required further investigation.
23.07.2026	I reviewed why the results were different between training runs and found that random seeds had not been properly set throughout the training process. This caused differences in model initialization and the order of training batches therefore identified the need to control the random settings to make the model results more consistent and reproducible.
26.07.2026	Conducted a controlled experiment by running the TCN model four times using the same configuration and dataset. Fresh checkpoint names were used for each run to ensure that previous model states did not influence the results then analysed the results from the four independent training runs. The Macro-F1 scores ranged from 0.756 to 0.840, with a mean of 0.788 and a standard deviation of 0.032. The variation demonstrated that selecting only the highest-performing run could provide a misleading representation of the model's performance. Therefore, I decided to use a more consistent selection approach based on the overall distribution of results.
27.07.2026	Finalized the Phase 1 TCN model by selecting the run with a Macro-F1 score closest to the overall mean (0.7854). The run achieving 0.840 was not selected because it represented the highest result in the observed distribution and could have introduced bias if reported as the representative performance. The selected model was saved as TCN_FINAL_layer2.keras.
30.07.2026	I examined the per-class classification results of the selected TCN model to identify classes requiring further improvement. The sign **"me"** showed the weakest performance, with a test recall of 0.00 and an F1-score of 0.00. The classification report also showed significant confusion between this class and other sign then I had to review the possible reasons for the poor performance of the **"me"** class by comparing its motion characteristics with those of the overall dataset. The average motion energy for this class was 0.9495, compared with the dataset average of 1.7887. This indicated that the sign contained relatively low movement and may have been more affected by the fixed-magnitude augmentation and derivative-based features.

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	After evaluating multiple training runs and selecting the model based on the mean performance rather than the highest individual score, the final TCN model achieved **0.79 test accuracy and 0.79 Macro-F1** across the 59 caregiver-specific classes using the 355-sample test set. These results were recorded as the baseline for the next stage of model development.

Supervisor's Comments

Student progressing well in research project: Yes / No
If No:

Project work to be carried out before next evaluation period:

Code base checked: Yes/No

Signature:

Date: 03.08.2026



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26 REGULAR RP

Research Project Periodic Log Entry Form

Project ID: R26-DS-010

Student ID: IT22178640

Name: Dayana Priyadharshani Kumar

Individual Component: AI Assisted Caregiver Workforce Deterioration Detection and Adaptive Workload Redistribution

Summary of RP work carried out during the reporting period

Date	RP Work
01.08.2026	Began the correction procedure for the lag1_stress scaling bug identified the previous day. Re-extracted EMA stress labels from the TILES-2018 source data to confirm the correct raw 1–5 scale values for all affected participants. Rebuilt the Stage 2 training dataset with lag1_stress represented consistently at the raw scale rather than binarized. Retrained the LGBMClassifier using the corrected feature and validated that class distributions and feature importances aligned with expectations. Compared preliminary retrained performance against the buggy version to confirm the fix addressed the degenerate prediction pattern. Documented the root cause and fix in preparation for backend redeployment.
02.08.2026	Focused on propagating the lag1_stress correction into the live Supabase database. Wrote a Colab-based update script to recompute and overwrite the affected lag1_stress values across all existing rows using the raw EMA scale. Validated the update by spot-checking a sample of participant records against their original EMA survey responses. Confirmed no downstream tables or views referenced the old binary encoding in a way that would reintroduce the bug. Reviewed Supabase RLS policies and GRANT statements on the affected tables to ensure the update script had the correct permissions to run. Logged the database correction as complete and ready for model artifact replacement.



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03.08.2026	Replaced the three production .joblib artifacts : baseline_regressor.joblib, lgbm_model_baseline.joblib, and lgbm_model_baseline_imputer.joblib with the corrected versions retrained on the fixed data. Re-ran the full test evaluation to confirm the corrected Stage 2 classifier achieved a test ROC-AUC of 0.7056. Verified the exact 16-feature input order expected by the imputer matched what the backend inference pipeline supplied. Re-tested the two-stage pipeline end-to-end, confirming hr_mean_deviation_model was correctly computed from Stage 1's output before being passed into Stage 2. Uploaded the corrected artifacts to the designated Google Drive cache location for team access. Confirmed the critical bug was fully resolved across data, model, and deployment layers.
04.08.2026	Investigated whether Bluetooth proximity network centrality contributed meaningful predictive value to the stress-risk model. Ran a three-model ablation comparing Random Forest, XGBoost, and LightGBM with and without proximity-based features included. Found no consistent improvement in performance across any of the three model types when proximity centrality was added. Cross-checked the ablation results against multiple evaluation runs to rule out data-splitting artifacts. Began drafting the reasoning for excluding proximity centrality as an active model input while retaining it as a documented research finding. Flagged the result as requiring independent statistical confirmation before finalizing.
05.08.2026	Ran an independent t-test to statistically confirm the null finding on proximity network centrality observed in the model ablation. Obtained a t-statistic of -1.52 and a p-value of 0.129, with an effect size of $d = -0.061$, confirming no significant association with stress outcomes. Documented the statistical test alongside the model-based ablation as converging evidence for the exclusion decision. Positioned the finding explicitly as "tested and excluded" rather than as an active feature, correcting an earlier framing error. Decided the result would be surfaced only in an admin-only research context rather than used to drive predictions. Prepared summary notes on this finding for later inclusion in the paper's methodology and limitations discussion.
06.08.2026	Shifted focus to identifying the strongest statistical predictors of stress within the TILES-2018 feature set. Ran correlation analysis on sleep efficiency against EMA stress outcomes and found a statistically significant negative association. Confirmed the result with $r = -0.106$ and $p < 0.0001$, marking it as the strongest single statistical finding in the dataset. Cross-validated the correlation across different participant subsets to check for stability. Flagged sleep efficiency as a feature to ensure was explicitly retained in the final model's feature summary. Began documenting this finding as a key contribution to the research narrative.



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07.08.2026	Began scaffolding the PULSE frontend dashboard using React 19, Vite, TypeScript, and Tailwind v4. Set up Zustand for state management and configured Supabase-js for data access from the frontend. Built the dark-gradient collapsible sidebar component intended for shared use across the CareSense platform. Established the tab-based module UI structure with consistent styling (44px height, 16px padding, 2px border, 14px font-size) intended for team-wide consistency. Documented these styling conventions in a shareable reference for the other component owners. Confirmed the sidebar and navigation structure did not conflict with files owned by teammates.
08.08.2026	Developed the Overview tab of the PULSE dashboard as the primary landing view for caregiver stress-risk information. Connected the tab to a backend endpoint returning summarized risk metrics across the caregiver pool. Implemented Recharts-based visualizations to present aggregate risk trends at a glance. Tested the tab against real TILES-2018-derived demo data to confirm accurate rendering. Adjusted layout and data formatting based on initial visual review. Verified the tab integrated cleanly with the shared sidebar and navigation components.
09.08.2026	Built the Risk Analysis tab to present individual caregiver stress-risk predictions from the two-stage LightGBM pipeline. Connected the tab to the corrected backend inference endpoint to display up-to-date predictions reflecting the lag1_stress bug fix. Implemented visual breakdowns of contributing factors using the finalized 16-feature set. Verified that displayed predictions matched the retrained model's outputs for the same input records. Handled edge cases where incomplete caregiver data required fallback display states. Cross-checked the tab's data flow against the Risk Analysis backend endpoint contract.
10.08.2026	Developed the Caregiver Profiles tab to present individual-level information for each caregiver in the demo pool. Integrated the fabricated ward field from caregiver_profiles for display and navigation purposes only, ensuring it was clearly separated from any logic feeding the redistribution model. Connected the tab to real participant-level attributes drawn from the TILES-2018 dataset where appropriate. Implemented profile-level historical prediction views to support later trend analysis. Verified that display-only fields could not inadvertently influence any backend calculations. Tested the tab against the 15-caregiver demo pool for consistency.
11.08.2026	Built the Trends tab to surface longitudinal patterns in caregiver stress-risk data. Included the proximity network centrality finding as an admin-only research result, explicitly labeled as a tested-and-excluded feature rather than an active model input. Implemented visualizations for the sleep efficiency correlation as a key highlighted finding. Structured the tab to separate empirical research findings from operational model outputs to avoid misrepresenting either. Reviewed the tab with the admin-only access restriction in mind to prevent premature exposure of unvalidated findings. Verified rendering accuracy against the underlying statistical results.



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12.08.2026	Developed the Redistribution tab to generate workload redistribution recommendations for caregivers. Sourced real Shift (Day/Night) and real PrimaryUnit (anonymized unit codes) fields from participant_info.csv.gz as the honest inputs driving the redistribution logic. Confirmed the fabricated ward field was excluded entirely from this logic, preserving the display/navigation-only boundary established earlier. Framed the tab's recommendations explicitly as design contributions rather than empirically validated interventions, given the absence of a live deployment context. Tested redistribution outputs against the demo caregiver pool for logical consistency. Documented the scoping rationale for later inclusion in the paper's limitations section.
13.08.2026	Built the admin-only Demo Control tab to manage demonstration scenarios for the PULSE dashboard. Implemented controls allowing switching between different demo caregiver states to showcase model behavior during presentations. Restricted access to this tab to admin accounts, consistent with the platform's admin-only account creation policy via Supabase Edge Function. Verified that demo data used in this tab remained local and excluded from version control per established data-handling practice. Tested the tab's controls against the 15-caregiver demo pool with real raw Fitbit and EMA files. Confirmed the tab did not expose any research-only findings outside their intended admin context.
14.08.2026	Implemented the five backend endpoints required to support the PULSE dashboard's tabs. Built endpoints using FastAPI, Python, and uvicorn, connected to the shared Supabase/PostgreSQL database. Added httpx keepalive configuration and an automatic retry helper to improve resilience against RemoteProtocolError failures. Tested each endpoint against the corrected two-stage model pipeline to confirm accurate, consistent responses. Verified endpoint response formats aligned with what each frontend tab expected. Documented endpoint contracts to support ongoing frontend-backend coordination.
15.08.2026	Reviewed and hardened Supabase security configuration across all new PULSE-related tables. Confirmed that both explicit RLS policies and explicit GRANT statements to authenticated and/or service_role were in place for every new table, addressing a class of issue that had previously caused real debugging incidents. Queried the live database to confirm the true highest migration version before creating any new migration files, avoiding the collision risk previously caused by uncommitted direct pushes. Documented these practices as a team-wide warning in the project README. Verified "Automatically expose new tables" remained disabled and automatic RLS remained enabled at the project level. Cross-checked Cloudflare R2 configuration for binary model artifact storage.

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26 REGULAR RP

Supervisor's Comments

Student progressing well in research project: Yes / No
If No:

Project work to be carried out before next evaluation period:

Code base checked: Yes/No

Signature:

Date: 20.08.2026



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26 REGULAR RP

Research Project Periodic Log Entry Form

Project ID: R26-DS-010

Student ID: IT22057488

Name: Harishalinee Elangovan

Individual Component: Edge-Enabled Skeletal Movement Analysis System for Real-Time Patient Observable Fall Risk Detection and Prediction

Summary of RP work carried out during the reporting period

Date	RP Work
01.08.2026	Model A training completed final epoch convergence reached with early stopping at patience 10. Recorded final metrics: Fusion achieved 98.27% accuracy, macro-F1 of 0.977, ROC-AUC of 0.993 on the NTU random-split test set (578 sequences). Generated eval_report.md with full per-class classification report, confusion matrix, and threshold value saved to threshold.json.
02.08.2026	Ran subject-disjoint evaluation holding out entire NTU subjects unseen during training. Fusion achieved 97.12% accuracy and macro-F1 of 0.961 — only 1.6% drop from the random split, confirming the model generalises to new subjects rather than memorising individuals. Saved eval_report_subj.md, stgcn_best_subj.pth, fusion_best_subj.pth, and threshold_subj.json.
03.08.2026	Conducted cross-dataset evaluation: NTU-trained model evaluated on all 70 UR Fall Detection clips. Fusion scored 42.86% accuracy (ROC-AUC 0.41), confirming the domain gap between Kinect world-metre 3D coordinates used in NTU and MediaPipe normalised 2D coordinates used in UR and the deployment webcam. Documented as a known limitation in eval_report_cross_dataset.md — directly motivates the webcam fine-tuning step planned in finetune_webcam.py.
04.08.2026	Designed the Supabase database schema for the SENTRY component. Wrote migration files 0007–0010 covering patients, rooms, fall_events, and fall_alerts tables with appropriate foreign keys, RLS policies using an is_admin() role helper, and identity primary keys. Verified all tables can be read and written from the Python backend using the service role key.

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05.08.2026	Implemented src/database.py replacing the earlier SQLite stub with a full Supabase client layer. Implemented all CRUD functions for patients, rooms, caregivers, fall events, and fall alerts. Applied HTTP/1.1 enforcement on the httpx session to prevent RemoteProtocolError on idle connection reuse that was crashing background async broadcast tasks.
06.08.2026	Implemented the FastAPI REST server in api/server.py with endpoints for patients, rooms, alerts, caregivers, dashboard summary, and camera configuration. Set up the lifespan handler for database initialisation and auto-start of inference engines for all rooms with a configured camera_src on server boot. Configured CORS middleware and tested all endpoints via the Swagger UI.
07.08.2026	Implemented WebSocket endpoint /ws/live/{room_id} broadcasting skeleton + risk data at ~5 Hz. Added a separate high-frequency skeleton-only path sending only joint coordinates at 25 FPS to reduce rendering latency on the client. Differentiated the two message types using a type field ("skeleton" vs full update) so clients can merge frames rather than replace full state on every message.
08.08.2026	Implemented src/postprocess.py risk state machine with EMA temporal smoothing (α=0.4). Defined state transitions with sustained-threshold logic: NORMAL→MODERATE requires score above 0.35 for 1.5 seconds; MODERATE→HIGH requires score above 0.65 for 0.5 seconds; HIGH→MODERATE requires score below 0.65 for 3 seconds. Added a 30-second HIGH alert cooldown preventing repeated alerts for the same event.
09.08.2026	Implemented the context engine in src/context_engine.py. Classifies posture (STANDING, SITTING, LYING, WALKING, TRANSITION, UNKNOWN) from joint geometry: torso inclination angle, knee-hip vertical drop, and ankle velocity. Implemented room zone polygon support — zones are stored as JSON in the rooms table and loaded at engine start to define safe and restricted areas. Zone membership modifies alert sensitivity thresholds.
10.08.2026	Implemented the inference engine in src/inference.py. Reads frames from a configured camera source (webcam index or RTSP URL), runs MediaPipe Pose per frame, maps 33 landmarks to the canonical 14 joints, and maintains a 90-frame sliding ring buffer at 30 FPS. Triggers model inference approximately 5 times per second by passing the buffered window through the ST-GCN, feature extractor, and fusion model. Tested stable operation over 60 minutes without memory growth.



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11.08.2026	Conducted recording session with Subject 2 (second volunteer). Recorded 15 fall clips covering forward, backward, and side falls on a yoga mat. Verified all 14 MediaPipe joints detected consistently throughout each sequence using the live overlay from record_clip.py built in July. This completes the two-subject webcam dataset (Subject 1: 35 clips recorded July 26, Subject 2: 15 fall clips today).
12.08.2026	Recorded Subject 2 normal activity clips (43 clips: walking, sitting, standing, reaching, turning, lying down). Ran extract_webcam_mediapipe.py on all Subject 2 clips confirming output shape (T, 14, 4). Re-ran parse_webcam.py to aggregate the complete 93-clip dataset into webcam_data.npy, webcam_features.npy, webcam_labels.npy, and webcam_subjects.npy. Executed finetune_webcam.py using Subject 2 as held-out validation (subject-disjoint split). Freeze-then-unfreeze strategy: backbone frozen for first 3 epochs then full fine-tuning at LR=1e-5 for 20 epochs. Best validation macro-F1 saved as stgcn_best_webcam.pth and fusion_best_webcam.pth.
13.08.2026	Built the React/Vite web dashboard. Implemented Login, Dashboard Overview, and Rooms tabs. Dashboard overview renders KPI stat cards (total alerts, HIGH/MODERATE counts, acknowledged today), a camera coverage donut chart derived live from rooms data, and an alert trend bar chart. Rooms tab lists all rooms with live/offline pill based on camera_src presence and expandable room detail including caregiver assignment and zone status.
14.08.2026	Implemented the Live Monitoring tab with a full-canvas skeleton renderer using HTML5 Canvas API. Bones coloured by current risk level (green/amber/red). Risk metric chips (score, posture, zone, confidence) update from WebSocket. Added automatic reconnect on disconnect. Implemented the Alerts Management tab with a sortable, filterable alert table, acknowledge action, risk level badges, and room code display.
15.08.2026	Completed remaining web tabs: Patient Details (history chart, alert timeline), Event Replay (skeleton animation from R2 replay frames), Analytics (risk distribution, alert frequency by room), Room Configuration (camera source input, patient assignment), Zone Configuration (polygon zone editor on canvas), and Users & Roles (caregiver-to-room assignment). Conducted end-to-end test across



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	all 11 tabs — confirmed backend connectivity, live skeleton streaming, and alert flow.

Supervisor's Comments

Student progressing well in research project: Yes / No
If No:

Project work to be carried out before next evaluation period:

Code base checked: Yes/No
Signature:
Date: 20.08.2026



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26 REGULAR RP

Research Project Periodic Log Entry Form

Project ID: R26-DS-010

Student ID: IT22294098

Name: Vidursha Prabagaran

Individual Component: Context-Aware Intelligent System for Automated ADL Extraction and Real-Time Documentation from Caregiver Voice Logs

Summary of RP work carried out during the reporting period

Date	RP Work
03.08.2026	Began Stage 3 (auto-editing) data preparation. Loaded and combined DisfluencySpeech and DISCO English (3,979 human-annotated pairs) into a unified raw/clean text schema, and constructed 25 hand-written, domain-grounded disfluency examples covering medication, meal, and hygiene phrasing, oversampled 5x to prevent dilution by the much larger general-purpose sources.
04.08.2026	Established zero-shot and rule-based baselines for the auto-editing stage, evaluating T5-small, BART-base, and a keyword-based filler-removal method on both the general test set and a held-out domain-specific test set, confirming neither zero-shot model yet outperformed the trivial rule-based baseline — motivating the fine-tuning step.
05.08.2026	Fine-tuned T5-small on the combined ~7,783-pair training set. Debugged and resolved a batch-padding error and a GPU/CPU device-placement error encountered during training, then confirmed a substantial improvement over baseline (general Exact Match rising from 25.6% to 67.2%).
07.08.2026	Fine-tuned BART-base on the same training set under identical conditions for a controlled architecture comparison. Diagnosed and corrected a text-parsing artifact (BART substituting the field separator with stray characters) that had been artificially depressing its exact-match score; corrected field-level results showed BART performing comparably to T5-small.



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08.08.2026	Conducted qualitative testing of both fine-tuned models on unseen sentences. Identified that T5-small failed to resolve a self-correction involving an elevated blood-pressure reading, retaining the incorrect pre-correction value, while BART-base resolved the same case correctly. Selected BART-base as the Stage 3 final model on this safety basis, despite comparable aggregate scores, establishing the safety-stratified selection methodology used through the remainder of the project.
09.08.2026	Implemented the Stage 4 grammar-correction component using a pretrained rule-based/ML tool, and constructed a manual test set of 15 caregiver-style sentences with deliberate grammar errors to evaluate its output ahead of pipeline integration
10.08.2026	Began Stage 5 (structured ADL extraction) dataset construction. Built a template-based data generator grounded in the care facility's real, anonymised medication regimens, weekly meal plan, routine descriptions, and documented symptom list, producing a 2,315-row dataset across nine ADL categories with patient identities replaced by anonymised codes (P01-P41) matching the facility's real gender and medication distribution.
11.08.2026	Established the rule-based baseline and fine-tuned T5-small and BART-base on the original six-field ADL schema. Both models substantially outperformed the baseline on field accuracy and alert recall; began qualitative safety testing on unseen sentences to inform final model selection.
13.08.2026	Completed qualitative testing for the six-field schema: identified that T5-small failed to flag a clearly elevated blood-pressure observation as alert-worthy, while BART-base correctly flagged it. Selected BART-base as the final model for this schema version, consistent with the safety-stratified selection principle established at Stage 3.
14.08.2026	Identified that the original six-field schema captured only whether an ADL activity occurred, not its specific content, and began designing an expanded 19-field schema capturing category-specific detail (e.g. specific food items, vital-sign readings, hygiene activity type) to strengthen the practical usefulness of extracted records.

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15.08.2026	Fine-tuned BART-base on the same expanded schema; identified degenerate generation behaviour and poor field-level accuracy on free-value fields (e.g. dosage, vital reading), with alert recall of only 42.6%. Attempted beam-search decoding as a first remediation step, which did not resolve the issue.

Supervisor's Comments

Student progressing well in research project: Yes / No
If No:

Project work to be carried out before next evaluation period:

Code base checked: Yes/No
Signature:
Date: 20.08.2026



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26 REGULAR RP

Research Project Periodic Log Entry Form

Project ID: R26-DS-010

Student ID: IT22289384

Name: Kaushalya Nagenthraraja

Individual Component: Personalized Sign Language Tutoring System for Caregivers

Summary of RP work carried out during the reporting period

Date	RP Work
01.08.2026	I started developing Layer 3, the Error-Localization Engine, with a focus on providing more detailed feedback on learner signing errors. I integrated Multivariate Dynamic Time Warping (DTW) using the dtw-python library and configured it with the symmetric2 step pattern. The system was designed to compare the original variable-length normalized sequences before resampling.
03.08.2026	Researched and identified the method **Medoid** and developed the reference selection method using a **Medoid** approach. The system identifies the training sample with the lowest overall DTW distance to the other samples within the same class. This sample is then used as the reference sequence for comparing learner gestures.
04.08.2026	Conducted initial checks to verify that the DTW calculations were working as expected. A same-class comparison, such as "sick" versus "sick", produced a distance of 1.99, while a comparison between different classes, "sick" versus "walk", produced a distance of 6.47. This confirmed that the system was able to distinguish between similar and different signs based on movement patterns.
05.08.2026	I analysed the landmark-level deviation results and identified an issue with the error-ranking process. Some naturally noisy landmarks, such as right_hand_8, were repeatedly identified as having large errors even when the learner performed the correct sign. This showed that using the raw spatial distance alone could produce misleading feedback.



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06.08.2026	Addressed the landmark deviation issue by introducing Z-score normalization. Instead of ranking errors only according to their raw distance, the system now considers how unusual the deviation is compared with the normal variation of each landmark. This was intended to provide more meaningful error rankings.
07.08.2026	Visually reviewed several reference sequences for the signs "mouth", "sick", and "pain" after noticing unusual DTW results. Although these sequences were shorter than the previously identified 213-frame threshold, their motion patterns contained multiple distinct movement peaks. This indicated that some reference samples may contain more than one sign repetition and could affect the accuracy of DTW comparisons.
08.08.2026	I conducted a wider dataset analysis to identify similar cases that were not detected by the previous 213-frame length threshold. Using motion-energy analysis on sequences between 90 and 200 frames, I identified 135 out of 2,364 samples (approximately 5.7%) as potentially containing multiple sign repetitions then refined the detection process by requiring the motion energy to remain low for a sufficient period before identifying a possible separation point. This helped reduce false detections caused by short natural pauses during signing. After applying the improved method, the number of potentially affected samples was reduced from 135 to 92. Higher proportions were observed in classes such as "stomach" (21%) and "drink" (11%).
09.08.2026	I manually reviewed a selection of the highest-confidence samples identified by the automated analysis. Two different movement patterns were observed: genuine concatenated signs, such as "eat" and "sit", and continuous repeated movements, such as "walk". This showed that not every repeated movement should automatically be treated as corrupted data.
12.08.2026	Based on the manual review, I decided not to automatically trim sequences containing continuous or cyclic movements. For signs such as "walk", repeated movement is part of the natural signing pattern, so automatically removing these sections could result in the loss of valid sign information then developed the is_true_concatenation classification method to distinguish genuine concatenated signs from continuous movement patterns. The method uses the sequence's own 10th-percentile motion-energy level as a reference for identifying meaningful rest periods before applying segmentation.

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13.08.2026	Developed the initial Stage 1.5 pre-filtering process for the Medoid reference selection. The purpose of this stage is to ensure that only verified single-sign sequences are included when calculating DTW distances. This helps prevent corrupted or concatenated samples from being selected as reference sequences.
14.08.2026	I reviewed the remaining tasks for Layer 3 and documented the areas requiring further work. The main pending activities include verifying the updated Z-score-based error rankings and completing the reference re-selection process for the affected "mouth", "sick", and "pain" classes. These tasks will be completed before progressing to the next layer of the system.

Supervisor's Comments

Student progressing well in research project: Yes / No

If No:

Project work to be carried out before next evaluation period:

Code base checked: Yes/No

Signature: 

Date: 20.08.2026



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26 REGULAR RP

Research Project Periodic Log Entry Form

Project ID: R26-DS-010

Student ID: IT22178640

Name: Dayana Priyadharshani Kumar

Individual Component: AI Assisted Caregiver Workforce Deterioration Detection and Adaptive Workload Redistribution

Summary of RP work carried out during the reporting period

Date	RP Work
16.08.2026	Drafted the methodology section for the ICAC 2026 paper covering the PULSE component. Documented the two-stage LightGBM pipeline architecture, including the 3-feature Stage 1 regressor and the 16-feature Stage 2 classifier. Wrote sections on dataset use, feature engineering, and the corrected lag1_stress handling reflecting the earlier production bug fix. Flagged internally that the SHAP explainability description referenced a discontinued earlier pipeline rather than the current TILES-based production model. Noted the proximity centrality result needed to be framed strictly as tested-and-excluded to avoid the earlier accuracy error. Identified the draft exceeded the target word count and required trimming.
17.08.2026	Shifted focus to preparing PULSE for cross-component integration with the rest of the CareSense platform. Reviewed shared files including Sidebar.tsx and navigation.ts to coordinate PULSE's integration without disrupting components owned by Harshu (SENTRY), Kaushalyaaa (Sign & Vitals), and Vidursha (SCRIBE). Verified that PULSE's five backend endpoints and dashboard tabs functioned correctly alongside the other components within the shared Supabase database. Pushed all PULSE work 82 files to the PULSE-01 branch with a clean push confirmed. Began preparing a pull request from PULSE-01 into the shared pp2-foundation branch. Coordinated initial end-to-end integration testing with teammates to confirm PULSE's outputs interoperated correctly with the other three modules ahead of the ICAC 2026 submission.



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18.08.2026	Continued cross-component integration testing with the team, focusing on verifying that PULSE's risk outputs did not conflict with data consumed by SENTRY's fall detection alerts. Tested shared Supabase table access patterns to confirm no read/write contention existed between PULSE and the other three modules. Reviewed navigation.ts changes proposed by teammates to ensure PULSE's tab routes remained intact after shared-file edits. Ran manual end-to-end walkthroughs of the dashboard with all four components active simultaneously. Logged minor UI inconsistencies at module boundaries for follow-up. Coordinated with Harshu to resolve a sidebar rendering conflict identified during joint testing.
19.08.2026	Audited the model versioning documentation to ensure the three production .joblib artifacts were clearly traceable to their corresponding training runs. Added version tags and training-date metadata to the Google Drive cache directory for baseline_regressor.joblib, lgbm_model_baseline.joblib, and lgbm_model_baseline_imputer.joblib. Cross-referenced each artifact against the Colab notebook cell outputs that produced it, to preserve reproducibility. Updated internal documentation describing which artifact version was currently live in production versus archived. Verified teammates had correct access permissions to the updated Drive structure. Flagged this as important groundwork ahead of final paper submission and code archiving.
20.08.2026	Drafted the results section reporting the corrected Stage 2 classifier's test ROC-AUC of 0.7056. Wrote up the sleep efficiency correlation finding ($r = -0.106$, $p < 0.0001$) as the strongest individual statistical result in the dataset. Presented the proximity network centrality ablation, reporting consistent null results across the Random Forest, XGBoost, and LightGBM comparisons. Included the independent t-test confirmation ($t = -1.52$, $p = 0.129$, $d = -0.061$) supporting the exclusion decision. Structured the section to separate model-performance results from standalone statistical findings. Reviewed the draft to ensure no metric was stated without its corresponding source.
21.08.2026	Drafted the related work section situating PULSE within existing literature on wearable-based caregiver and workforce stress detection. Compared PULSE's causally-ordered, two-stage design against single-stage classification approaches common in prior stress-detection studies. Summarized several closely related studies to establish context for the novelty statement. Positioned PULSE's use of a personalized baseline-deviation feature as a point of differentiation from prior work. Cited the TILES-2018 dataset paper (Mundnich et al., Sci. Data, 2020) as the mandatory dataset citation within this section. Reviewed citation formatting for IEEE compliance.



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22.08.2026	Drafted the novelty statement summarizing PULSE's core contribution: a personalized, causally-ordered stress-risk detection approach built on a validated two-stage pipeline. Articulated how the baseline-deviation design differs from directly modeling raw physiological values. Highlighted the sleep efficiency finding and the transparently reported proximity-centrality null result as evidence of methodological rigor rather than only positive results. Cross-checked the novelty claims against the actual tested and validated components of the pipeline to avoid overstatement. Revised phrasing for conciseness. Circulated the draft internally for consistency with the shared Introduction section.
23.08.2026	Drafted the integration contribution section describing how PULSE fits within the broader CareSense PP2 platform alongside SENTRY, Sign & Vitals, and SCRIBE. Described PULSE's five backend endpoints and how they were designed to interoperate with the shared Supabase database used by all four components. Explained the shared UI conventions (sidebar, tab structure, styling standards) established to maintain consistency across modules. Documented how PULSE's outputs were intended to complement fall-detection and vitals data from the other components in a unified caregiver-monitoring context. Reviewed the section against actual integration testing outcomes to ensure accuracy.
24.08.2026	Drafted the limitations section covering PULSE's scope and constraints. Documented the exclusion of proximity network centrality as a tested-but-unused feature, framed accurately as a null finding rather than a flaw. Noted the reliance on TILES-2018's historical data to simulate application inputs rather than live wearable streams, and the resulting distinction between research experimentation and real-world deployment. Scoped workload redistribution recommendations explicitly as a design contribution rather than an empirically validated intervention, given the absence of a live deployment context. Reviewed the section to ensure limitations were stated candidly without undermining the reported results.
25.08.2026	Revised the methodology section after re-reading it against the actual corrected model behavior, ensuring the lag1_stress feature was described as using the raw EMA 1–5 scale throughout, consistent with the earlier production bug fix. Verified the final feature summary explicitly included sleep efficiency and workload alongside the previously listed physiological features. Corrected the SHAP explainability description to reference the current TILES-based production model rather than the discontinued earlier pipeline. Rewrote the proximity-centrality description in the results section to avoid overstating the null result. Trimmed the combined methodology and results text toward the paper's word-count target.



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26.08.2026	Conducted a full internal review of all PULSE-owned sections — methodology, results, related work, novelty statement, integration contribution, and limitations — for factual and numerical consistency. Cross-checked every reported figure (ROC-AUC, correlation, t-test statistics) against its original source computation. Verified citation accuracy for both the TILES-2018 dataset paper and the SHAP explainability reference. Reviewed section flow to ensure the writing avoided reconstructing any external source's structure or wording. Finalized the PULSE-owned sections as ready for integration with the team's shared Abstract, Introduction, Architecture, Discussion, and Conclusion sections.
27.08.2026	Sent the completed draft of the ICAC 2026 paper, including all PULSE-owned sections, to the main supervisor and co-supervisor for review. Received marked-up feedback the same day covering both PULSE-specific sections and the shared sections of the paper. Reviewed the supervisors' comments on the methodology and results sections, noting requests for clearer justification and minor wording corrections. Went through feedback on the related work and limitations sections to identify which points required rewriting versus light editing. Consolidated all comments into a single tracked list to organize the correction work planned for the following day. Coordinated briefly with teammates to confirm which sections each person would be responsible for correcting.
28.08.2026	Worked through the supervisors' marked-up comments to correct the PULSE-owned sections of the paper. Rewrote flagged sentences to address clarity and accuracy concerns while keeping the overall structure and tone intact. Humanized sections of the text that had been flagged as reading too mechanically, adjusting sentence flow and word choice without altering the underlying technical content. Ran the revised paper through Turnitin to generate a similarity report ahead of submission. Reviewed the similarity report results and confirmed no problematic overlap existed in the PULSE-owned sections. Circulated the corrected and checked draft back to the team for a final look.
29.08.2026	Completed the full end-to-end workflow for the demo pool used in PULSE's dashboard demonstrations. Verified that all 15 demo-pool caregivers' raw Fitbit, daily summary, and EMA data flowed correctly through preprocessing, inference, and display without errors. Finalized the workload redistribution workflow, confirming real Shift and PrimaryUnit fields from participant_info.csv.gz correctly drove redistribution outputs end-to-end. Tested the redistribution logic against the full demo pool to confirm recommendations were generated consistently across different caregiver scenarios. Verified the fabricated ward field remained correctly excluded from redistribution calculations throughout the completed workflow. Logged both the demo pool and redistribution workflows as fully functional and ready for final integration.

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30.08.2026	Completed UI enhancements across the PULSE dashboard, refining layout, spacing, and visual consistency across all tabs ahead of final submission. Opened the pull request from PULSE-01 into the shared pp2-foundation branch with the enhanced UI and completed workflows included. Integrated PULSE with the rest of the CareSense platform and re-ran cross-component checks to confirm stability after merge. Checked the fall detection component (SENTRY) alongside PULSE to confirm no conflicts existed between the two modules' shared UI elements and Supabase table access. Verified the integrated dashboard rendered correctly with all four components active together. Confirmed the PR and integration work were complete and ready for the team's final review.
31.08.2026	Submitted the finalized CareSense conference paper, including all PULSE-owned sections, to the SLIIT ICAC 2026 conference following the completed corrections, humanization pass, and Turnitin similarity check. Confirmed the submission was successfully received through the conference's submission portal. Verified the submitted PDF reflected the final corrected methodology, results, related work, novelty statement, integration contribution, and limitations sections.

Supervisor's Comments

Student progressing well in research project: Yes / No

If No:

Project work to be carried out before next evaluation period:

Code base checked: Yes/No

Signature:

Date: 02.09.2026



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26 REGULAR RP

Research Project Periodic Log Entry Form

Project ID: R26-DS-010

Student ID: IT22057488

Name: Harishalinee Elangovan

Individual Component: Edge-Enabled Skeletal Movement Analysis System for Real-Time Patient Observable Fall Risk Detection and Prediction

Summary of RP work carried out during the reporting period

Date	RP Work
16.08.2026	Began Flutter mobile application for the SENTRY caregiver interface. Set up Riverpod state management, Supabase Flutter client, and Dio HTTP client. Implemented authentication screens (login, OTP, signup, onboarding, splash) in light theme. Fixed corrupted Unicode/emoji sequences causing compile errors inherited from earlier scaffold code.
17.08.2026	Implemented SENTRY home screen with caregiver-scoped stat cards. Room scoping logic: resolves logged-in user's auth UUID → profiles.name → caregiver_profiles.display_name → rooms.id (bigint FK), then filters alert counts to that room's numeric ID. Unacknowledged HIGH, MODERATE, LOW counts and total monitored patients displayed.
18.08.2026	Implemented Alerts screen with gradient hero design. Diagnosed and fixed alert count showing 0 on mobile: filter was comparing room_code string against room_id bigint — never matched. Fixed by selecting rooms.id bigint for comparison. Also included NULL room_id alerts (valid alerts where backend could not resolve room at insert time) to fix count discrepancy of 6 alerts between web (25) and mobile (19).
19.08.2026	Implemented Live Room screen with full-canvas skeleton CustomPaint. Diagnosed metric chips showing "- -" despite WebSocket active: backend sends skeleton-only frames (type="skeleton") at 25 FPS that were replacing the entire state and clearing risk metrics. Fixed using frame-merge logic: skeleton-only frames update only the skeleton key while preserving existing risk_score, posture, zone, and confidence values. Wrapped CustomPaint in RepaintBoundary for isolated repaints.



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20.08.2026	Implemented Patient Detail and Event Replay screens with real API data. Implemented Rooms screen filtered to caregiver's assigned rooms only. Completed Profile screen removing the About section. Removed all remaining hardcoded placeholder data across all SENTRY screens replacing with live SentryService API calls.
21.08.2026	Fixed acknowledge sync between mobile and web. The Supabase direct-write fallback in acknowledgeAlert() was writing to a non-existent acknowledged_at field. Corrected to write the three correct schema fields: acknowledged (boolean), ack_by (text), ack_at (timestamptz). Acknowledgements on mobile now reflect immediately on the web supervisor dashboard.
22.08.2026	Fixed alert cards and home screen recent alerts showing "Room 7" instead of "ROOM_04". Root cause: rooms.id starts at 4 due to seed order (ROOM_04 maps to id=7). Fix applied at backend: get_alerts() now joins the rooms table using select("*, rooms(room_code)") and flattens room_code to a top-level field. Mobile reads room_code from the response directly, eliminating raw numeric ID display.
23.08.2026	Added Supabase Realtime subscription to the Alerts screen using onPostgresChanges for INSERT and UPDATE events on the fall_alerts table. New alerts generated by the web backend now push to mobile instantly without polling. Implemented proper channel disposal in dispose() to prevent memory leaks. Verified new alert appears on mobile within one second of backend insertion.
24.08.2026	Wrote the research paper methodology section: system architecture, dataset pipeline (NTU RGB+D selection rationale, UR as external test only, in-house webcam for domain adaptation), ST-GCN graph design, 18 kinematic features rationale, late fusion architecture, and training protocol (class-weighted loss, val-optimal threshold, subject-disjoint splits).
25.08.2026	Wrote the research paper results section. Compiled the Model A three-way comparison table (fusion vs ST-GCN-only vs feature-only) across all three evaluation protocols. Wrote ablation study analysis. Added Model C webcam results: fusion 91.4% accuracy, 80% fall recall, 4.51 ms inference latency on CPU. Wrote limitations section covering the two-subject dataset size and the NTU-to-webcam domain gap.
26.08.2026	Ran the corrected webcam held-out evaluation (eval_webcam_holdout.py) on Subject 2's 58 clips — prior eval_report_webcam.md had incorrectly evaluated on the NTU test set (578 samples) rather than the webcam held-out set. New results:



	Fusion 91.4% accuracy, macro-F1 0.885, fall recall 80%, ROC-AUC 0.853. Updated report files and integrated results into the paper.
27.08.2026	Resolved merge conflict between SENTRY-03 branch and pp2-foundation on frontend/vite.config.ts. SENTRY change (historyApiFallback: true for SPA route refresh) and SCRIBE team change (proxy for /api/scribe) were in the same server block. Merged both settings as they operate on independent request paths. Pushed resolved branch and confirmed PR shows able to merge on GitHub.
28.08.2026	Prepared PP2 presentation slides. Structured as: clinical problem and gap, system architecture diagram (camera → MediaPipe → dual-branch model → post-processing → web + mobile), dataset decisions, training results table, webcam deployment results, live demo walkthrough plan, and commercialisation section. Reviewed slide content against PP2 marking rubric to ensure all LO criteria are addressed.
29.08.2026	Recorded demo video for PP2 presentation showing end-to-end system operation: room configuration on web supervisor dashboard, live skeleton canvas with updating metric chips, alert generation and acknowledgement on web, and Realtime alert push appearing on mobile within one second. Narrated inference latency (4.51 ms) and fusion accuracy (91.4%) during the recording.
30.08.2026	Submitted research paper to the target conference. Finalised formatting to conference template requirements including figure captions, reference list, and word count compliance. Performed final proofreading pass. Submitted through the conference management portal and received submission confirmation. Marked all Phase 10 research paper tasks as DONE in the Component Completion tracker.
31.08.2026	Created GitHub pull requests for the SENTRY component — SENTRY-03 into pp2-foundation (backend, web frontend, Supabase migrations) and sentry-mobile branch (Flutter caregiver app, sm-fix23 through sm-fix42) prepared for team review. Conducted final 1-hour end-to-end stability test confirming no crashes, no memory leaks, WebSocket maintained continuously, and alert acknowledge



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	syncing correctly between web and mobile. Updated bi-weekly log and finalised documentation for PP2 viva.

Supervisor's Comments

Student progressing well in research project: Yes / No
If No:

Project work to be carried out before next evaluation period:

Code base checked: Yes/No
Signature:
Date: 02.09.2026



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26 REGULAR RP

Research Project Periodic Log Entry Form

Project ID: R26-DS-010

Student ID: IT22294098

Name: Vidursha Prabagaran

Individual Component: Context-Aware Intelligent System for Automated ADL Extraction and Real-Time Documentation from Caregiver Voice Logs

Summary of RP work carried out during the reporting period

22.08.2026	Retrained BART-base for an extended 12 epochs to rule out under-convergence. Alert recall improved to 70.4%, but free-value field accuracy remained poor (20-30%), despite healthy-looking validation loss throughout training — identified and documented this as a methodologically significant finding: aggregate loss did not predict field-level extraction accuracy for this structured, heterogeneous-output task.
23.08.2026	Selected T5-small as the final model for the expanded schema on the basis of a clear, three-times-confirmed capability advantage over BART-base, contrasting with the Stage 3 and original-schema decisions where model choice depended on a narrow safety trade-off rather than a broad capability gap.
24.08.2026	Conducted extended qualitative error analysis of the final T5-small model across 52 hand-constructed sentences in three rounds, identifying and documenting two concrete limitations: systematic false-positive alerting on positively-framed observations, and one high-consequence miss in which a documented symptom (dizziness) was misclassified into the mobility field with its alert not raised.and

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26.08.2026	Implemented Stage 6 (template-based report generation) and Stage 7 (natural-language summary generation plus text-to-speech handover audio using an Indian English voice), and connected both to the Stage 5 output schema.
27.08.2026	I worked on web development, including both the frontend and backend.
29.08.2026	I worked on web development, including both the frontend and backend.
30.08.2026	I worked on web development, including both the frontend and backend.

Supervisor's Comments

Student progressing well in research project: Yes / No
If No:

Project work to be carried out before next evaluation period:

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26 REGULAR RP

Code base checked: Yes/No

Signature: 

Date: 02.09.2026



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26 REGULAR RP

Research Project Periodic Log Entry Form

Project ID: R26-DS-010

Student ID: IT22289384

Name: Kaushalya Nagenthraraja

Individual Component: Personalized Sign Language Tutoring System for Caregivers

Summary of RP work carried out during the reporting period

Date	RP Work
15.08.2026	Reviewed the current state of the GLOSS backend end-to-end against the original implementation plan to identify what remained before the system could be considered complete. Confirmed that the trained model, feature scaler, class thresholds, landmark ordering, and reference exemplars were all present and internally consistent with the original research specification. Identified two blocking architectural issues that needed to be resolved before further backend work could safely proceed: (1) the inference-time preprocessing was only exposing the resampled 60-frame representation, not the original variable-length sequence needed for execution-quality comparison, and (2) a dependency conflict between MediaPipe and TensorFlow meant both libraries could not be safely installed in the same Python environment.
16.08.2026	Resolved the environment conflict by isolating the MediaPipe-dependent landmark extraction into a standalone subprocess worker, invoked from the main FastAPI service via a lightweight bridge module. This keeps TensorFlow and MediaPipe in separate virtual environments while still allowing a single inference request to use both. Verified the full pipeline end-to-end from the main service process, confirming raw landmark extraction, preprocessing, and model prediction all completed correctly through the new subprocess boundary.
17.08.2026	Replaced the pickled scikit-learn LabelEncoder used to resolve class indices to sign names with a plain JSON class list, removing an unnecessary scikit-learn dependency and an sklearn version-mismatch warning at runtime. Implemented and tested the sign-recognition endpoint (video in, predicted sign and confidence out),



	including a corrupted-input failure path returning a proper 400 response rather than a server error.
18.08.2026	Located the original Google Colab notebook containing the training-time preprocessing pipeline and transcribed the landmark extraction, missing-landmark interpolation, per-frame body-anchored normalisation, temporal resampling, and kinematic feature generation steps directly from that source rather than reimplementing them from the written specification, to avoid any silent mismatch between training and inference preprocessing. Verified the transcribed pipeline against two real recorded videos, confirming plausible predictions consistent with known model behaviour.
19.08.2026	Attempted to locate the original DTW execution-scoring implementation referenced in earlier project documentation. The corresponding notebook could not be located. Reconstructed the underlying distance calculation instead by testing candidate formulations directly against the stored calibration evidence (the saved reference exemplars and calibration distance records). Confirmed that a standard symmetric-step dynamic time warping distance, computed over the full landmark feature set, reproduced the stored calibration values to within a small margin across several independently tested sign classes, and adopted this as the verified execution-scoring formula.
20.08.2026	Implemented the execution-quality evaluator using the reconstructed DTW formula, comparing a caregiver's attempt against the stored reference sequence for the target sign and classifying the result as strong, moderate, or weak against the existing per-class thresholds. Extended the DTW module to additionally return the frame-alignment path, not only the final distance, as a foundation for landmark-level feedback.
21.08.2026	Built the corrective-feedback module. Grouped the tracked landmarks into nine interpretable body regions and used the DTW alignment path to identify which regions deviated most from the reference sequence, generating a short plain-language summary for moderate and weak attempts while keeping the wording appropriately hedged rather than diagnostic.
22.08.2026	Designed and implemented the caregiver mastery-tracking engine. Streak progress is attempt-based rather than date-based, so infrequent practice is not penalised. Mastery status is recomputed from attempt counters using a fixed rule set (new, learning, weak, improving, mastered, needs revision), with an explicit requirement that a sign cannot be marked mastered until at least one webcam attempt has been independently verified as strong, so multiple-choice practice alone cannot fully substitute for demonstrated execution.



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23.08.2026	Implemented the adaptive lesson-selection engine, using weighted-random selection that favours signs with weaker recorded mastery. Added a separate, deterministic rule for a caregiver's first-ever lesson, based on population-level sign difficulty rather than personal history, since no personal history yet exists at that point.
24.08.2026	Combined recognition, execution evaluation, feedback generation, mastery updates, and lesson selection into a single attempt-submission endpoint. This endpoint requires a verified authenticated session rather than accepting a caller-supplied caregiver identity, so an attempt can only ever be recorded against the caregiver who is actually logged in.
25.08.2026	Extended the API with supporting read endpoints (sign catalogue, next-recommended-lesson) needed by the frontend. Before beginning frontend implementation, audited the existing shared frontend codebase - routing, authentication, and design conventions - to ensure the new GLOSS interface would integrate without modifying files shared with other components.
26.08.2026	Implemented the caregiver-facing GLOSS learning interface: personalised and browsable sign selection, webcam capture with manual start/stop recording, a multiple-choice fallback for when camera access is unavailable, and a results view showing the recognised sign, execution score, quality tier, corrective feedback, and updated mastery status.
27.08.2026	Began development of a live, data-driven avatar to demonstrate each sign, implemented as a procedural rig driven directly by the stored reference landmark sequences rather than pre-recorded video, so that the same underlying rig logic can later be applied to a licensed 3D character model. Also implemented a smaller, related caregiver-education feature covering common Parkinson's movement symptoms, reusing the same authentication and data-storage conventions as the sign-language component.
28.08.2026	Diagnosed and corrected a rendering fault in the avatar rig, in which a duplicated positioning offset moved the arm hierarchy outside the visible camera frame. Investigated a reported drop in live recognition accuracy on the deployed interface; confirmed through direct comparison testing that the browser's video recording format was not the cause, narrowing the remaining cause to known model-level recognition variability rather than an integration fault.
29.08.2026	Reviewed remaining open items for the next reporting period, including extending avatar verification beyond the initial proof-of-concept set of signs, adding a caregiver progress-history view, and sourcing reference media for the multiple-choice fallback. Documented these as pending work rather than treating the component as finished.

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31.08.2026	Submitted the finalized CareSense conference paper, covering all four components including GLOSS, to the SLIIT ICAC 2026 conference through the official submission portal. Confirmed the submitted PDF reflected the fully corrected GLOSS methodology, verified results, and finalized limitations and novelty statement following the supervisor review and Turnitin similarity check completed the previous days.

Supervisor's Comments

Student progressing well in research project: Yes / No
If No:

Project work to be carried out before next evaluation period:

Code base checked: Yes/No

Signature: 

Date: 02.09.2026



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26 REGULAR RP

Research Project Periodic Log Entry Form

Project ID: R26-DS-010

Student ID: IT22178640

Name: Dayana Priyadharshani Kumar

Individual Component: AI Assisted Caregiver Workforce Deterioration Detection and Adaptive Workload Redistribution

Summary of RP work carried out during the reporting period

Date	RP Work
01.09.2026	Participated in Progress Presentation 2 and presented the PULSE component within the integrated CareSense research project. Explained the two-stage stress-risk pipeline, personalized heart-rate baseline, model evaluation and supervisor-facing workload redistribution workflow. Discussed the distinction between a predicted elevated-stress response and a clinical burnout diagnosis. Also confirmed submission of the group research paper to ICAC 2026 by its 1st September deadline.
02.09.2026	Reviewed the feedback and questions raised during Progress Presentation 2. Rechecked how the PULSE demonstration presented historical TILES readings, risk probabilities and SHAP factors. Identified descriptions that needed clearer separation between research evaluation, application demonstration and actual care-home deployment.
03.09.2026	Revisited the PULSE research notebook to trace the participant-level separation used for training, validation and testing. Reviewed the saved performance outputs and the interpretation of the learned expected-heart-rate baseline. Checked that the written results did not overstate the model as a future burnout forecast.
04.09.2026	Worked on the PULSE evaluation narrative, comparing the full model with the previous stress response as a baseline. Reviewed the limitations of using lagged EMA responses as an input and the effect on claims about predicting stress when caregivers do not answer surveys.
05.09.2026	Reviewed the proximity-network experiments and the ablation results. Documented the absence of a detectable improvement from proximity centrality and ensured that this negative finding remained part of the research contribution rather than being presented as a successful predictive feature.
07.09.2026	Coordinated the research paper content across the four CareSense components. Reviewed the common description of PULSE, SENTRY, SCRIBE and GLOSS and checked that their individual evaluation results were distinguished from integrated platform functionality.

08.09.2026	Participated in the group research-paper writing session. Worked on aligning the paper structure, methodology explanations and reported component results. Reviewed the wording of the PULSE contribution and the boundaries of the available evaluation evidence.
09.09.2026	Continued revising the conference manuscript, focusing on the PULSE methodology and reported results. Cross-checked the model architecture, participant-grouped evaluation and limitations against the notebook and removed claims that could not be supported by the experiments.
10.09.2026	Reviewed the manuscript presentation of SHAP explanations and the workload redistribution demonstration. Clarified that the application displays directional contributing factors and that redistribution suggestions require supervisor review; no real-world stress reduction was measured.
11.09.2026	Checked the integrated application descriptions used in the paper, including the shared sign-in, role-based access and four component modules. Worked with the group to keep the common architecture description consistent with the actual application.
12.09.2026	Reviewed the evidence required for the revised conference paper and organized supporting research results. Focused on the held-out caregiver evaluation, uncertainty reporting and repeated participant-grouped cross-validation as evidence for the PULSE claims.
14.09.2026	Edited research documentation to distinguish the original proposal objectives from the experiments and functionality ultimately completed. Noted the limitations of the TILES hospital-worker population when discussing use in residential care homes.
15.09.2026	Prepared the PULSE and group research materials for the next round of manuscript revision. Checked consistency of the research title, author information, terminology, results and limitations across the working documents.

Supervisor's Comments

Student progressing well in research project: Yes / No
If No:

Project work to be carried out before next evaluation period:

Code base checked: Yes/No

Signature: 

Date: 18.09.2026



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26 REGULAR RP

Research Project Periodic Log Entry Form

Project ID: R26-DS-010

Student ID: IT22057488

Name: Harishalinee Elangovan

Individual Component: Edge-Enabled Skeletal Movement Analysis System for Real-Time Patient Observable Fall Risk Detection and Prediction

Summary of RP work carried out during the reporting period

Date	RP Work
01.09.2026	Reviewed PP2 supervisor comment ("improve model accuracy when environment changes"). Identified three existing code-level mechanisms that address it: Z-drop augmentation in data_processor.py (randomly zeroes depth channel, simulating sensor noise and occlusion), clip-level joint normalisation (removes camera-height bias so model generalises across room setups), and the fine-tuning pipeline in finetune_webcam.py (domain adaptation with new-environment clips). Prepared written justification for viva response — no code changes required.
02.09.2026	Began individual component thesis document. Created full chapter structure. Drafted abstract: 3,847 sequences (NTU RGB+D 3,777 + UR Fall Detection 70), 93 in-house webcam clips, subject-disjoint AUC 0.9836, webcam fine-tuned AUC 0.8527, end-to-end latency 5.55 ms, 49 automated tests.
03.09.2026	Wrote thesis Chapter 1 (Introduction): clinical motivation, fall incidence statistics in elderly populations, limitations of wearable and vision-based prior systems, SENTRY design goals (edge-deployable, skeleton-only, no retained video). Defined scope of individual component clearly against group system boundary.
04.09.2026	Wrote thesis Chapter 2 (Literature Review): pose estimation approaches (OpenPose, MediaPipe Pose), skeleton-based action recognition (ST-GCN variants), fall detection dataset landscape, edge deployment constraints. Identified gap: no prior work combines ST-GCN and handcrafted biomechanical features for webcam-only edge fall detection.



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05.09.2026	Wrote thesis Chapter 3 Sections 3.1–3.3: dataset selection rationale, NTU RGB+D action classes (A043 fall=947, A008=948, A009=948, A027=934), UR Fall Detection as external generalisation test only, in-house webcam for domain adaptation. Dataset Table 3.1 and NTU breakdown Table 3.2 drafted with verified counts.
06.09.2026	Wrote thesis Chapter 3 Sections 3.4–3.6: preprocessing pipeline, 7-step augmentation chain in confirmed order (temporal jitter, subwindow crop, LR flip p=0.5, yaw rotation, perspective+roll, Z-drop p=0.5, Gaussian noise). Section 3.6 ethics: stated form submitted, no institutional reply received, not claiming approval or exemption.
07.09.2026	Travelled to Elder Care Home for real-environment validation session. Three caregiver volunteers observed system on laptop. Recorded 8 simulated fall scenarios (sideways and forwards) and 12 normal activity sequences in the care home corridor. Obtained verbal consent from all participants for non-recorded observation (camera preview only, no video saved).
08.09.2026	Processed elder home validation results. System produced 7 HIGH alerts from 8 fall scenarios (87.5% fall recall) and 1 false HIGH alert from 12 normal sequences. SITTING cap correctly suppressed 2 alerts in chair-seated normal activities. Average latency 5.89 ms on care home device (AMD Ryzen 5), slightly above Ryzen 7 benchmark of 5.55 ms. Documented results for thesis qualitative section.
09.09.2026	Wrote thesis Chapter 4 Sections 4.1–4.3: system architecture overview, MediaPipe Pose joint extraction (14 of 33 landmarks), ST-GCN architecture (3 blocks: 4→32→64→128, temporal kernel=9, symmetric-normalised adjacency). LateFusionNetwork: LayerNorm(128) concat 18 features = 146-D → Linear(64) → BN → ReLU → Dropout(0.3) → Linear(2).
10.09.2026	Wrote thesis Chapter 4 Section 4.4: 18 biomechanical features (F1–F18) documented individually. F17 defined correctly as skeleton bounding-box height ratio (min/max across all 14 joints), with note that settings.py variable min_hip_height_ratio is a misnomer. F13 hip-knee angle weights (0.6/0.4) noted as judgement-fixed with no grid search.
11.09.2026	Wrote thesis Chapter 4 Section 4.5: post-processing pipeline. EMA α=0.25, τ_low=0.55, τ_high=0.72, ALERT_COOLDOWN_S=20.0, ALERT_MODERATE_COOLDOWN_S=120.0, 25-window dual-gate counter. ContextEngine: 6 posture states (STANDING, SITTING, LYING, WALKING,



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	TRANSITION, UNKNOWN). SITTING cap: HIGH→MODERATE when posture=SITTING. Zone modifier: computed but deliberately not applied to score.
12.09.2026	Wrote thesis Chapter 5: web system. Documented 10-tab web supervisor dashboard (dashboard, rooms, live, alerts, replay, history, analytics, reports, config, zoneconfig). Zone configuration uses hardcoded ZONE_PRESETS, not free-form polygon drawing. Exports: CSV and JSON only. No backend authentication on any endpoint — documented as known limitation.
13.09.2026	Wrote thesis Chapter 5 continued: Flutter mobile application. Screens documented (home, alerts with Supabase Realtime subscription, live room with skeleton CustomPaint, patient detail, event replay, rooms, profile). Room scoping logic: auth UUID → profiles.name → caregiver_profiles.display_name → rooms.id (bigint). Acknowledge sync fields: acknowledged (boolean), ack_by (text), ack_at (timestamptz).
14.09.2026	Wrote thesis Chapter 6 (Results). Table 6.1 random-split: ST-GCN AUC 0.9886, Feature-only AUC 0.9865, Fusion AUC 0.9926 (FAR 0.70%), RF AUC 0.9970 (accuracy 97.92%, confusion matrix [[426,5],[7,140]] summing to 578 — confirmed from rf_baseline_results.json and included in table). Subject-disjoint: Fusion AUC 0.9836, RF AUC 0.9882. Cross-dataset NTU→UR: Fusion AUC 0.410. Webcam: AUC 0.8527. Latency: 5.551 ms, 180.1 windows/s.
15.09.2026	Contributed SENTRY architecture section to group common thesis Chapter 4. Described edge deployment design, Supabase integration model, and multi-tenant room scoping. Attended group coordination meeting to align Chapter 4 contributions across SENTRY, SCRIBE, and GLOSS components. Confirmed section boundaries and no overlap with teammates' contributions.

Supervisor's Comments

Student progressing well in research project: Yes / No
If No:

Project work to be carried out before next evaluation period:

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Code base checked: Yes/No

Signature:

Date: 18.09.2026



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Research Project Periodic Log Entry Form

Project ID: R26-DS-010

Student ID: IT22294098

Name: Vidursha Prabagaran

Individual Component: Context-Aware Intelligent System for Automated ADL Extraction and Real-Time Documentation from Caregiver Voice Logs

Summary of RP work carried out during the reporting period

Date	RP Work
01.09.2026	I reviewed the outcome of the PP02 evaluation. The written feedback stated "No comments", but during the panel evaluation the panel asked me how SCRIBE would handle longer voice logs containing several sentences. Following this question, I tested the existing pipeline with a longer recording describing three activities and found that it produced only one empty medication record instead of separate records for each activity.
02.09.2026	I investigated the cause of the long-recording failure and found two main reasons. Whisper processes only the first 30 seconds of audio, and the T5-small extraction model was trained on one observation per input. Based on this, I planned to add audio chunking and sentence splitting to the pipeline.
03.09.2026	I implemented audio chunking, which splits recordings longer than 30 seconds into the smallest number of equal pieces of up to 30 seconds each. During testing, I found that fixed 30-second pieces left a very short final piece that produced a stray character, so I changed the method to use equal pieces and added character filtering.
04.09.2026	I implemented sentence splitting after transcription, so that each sentence is processed separately through BART-base, LanguageTool and T5-small. I then tested the updated pipeline with sample paragraphs containing several observations.

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13.09.2026	I built a multi-observation evaluation set of 100 paragraphs by joining two or three sentences from the ADL test split. The set contained 243 events, 64 of which required an alert, and was used to measure how well the pipeline handles several observations in one input.
14.09.2026	I ran the multi-observation evaluation and analysed the results. Sentence splitting raised event recall from 41.2% to 100% and alert recall from 18.8% to 100%, which confirmed that the pipeline could now recover every observation in a paragraph.
15.09.2026	I recorded a 30.8-second test recording with six caregiver observations, spoken naturally in simple English, and compared the old and new pipelines. The new pipeline produced separate records, but the test also showed speech recognition errors on important words, such as a medication name that was transcribed incorrectly and then changed further by LanguageTool.

Supervisor's Comments

Student progressing well in research project: Yes / No
If No:

Project work to be carried out before next evaluation period:

Code base checked: Yes/No
Signature:
Date: 18.09.2026



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Research Project Periodic Log Entry Form

Project ID: R26-DS-010

Student ID: IT22289384

Name: Kaushalya Nagenthraraja

Individual Component: Personalized Sign Language Tutoring System for Caregivers

Summary of RP work carried out during the reporting period

Date	RP Work
01.09.2026	Presented GLOSS at Progress Presentation 2 (PP2) and demonstrated the working system live: webcam practice of care signs, sign recognition, movement grading with DTW, corrective feedback, mastery tracking and the next recommended lesson, all stored in Supabase.
03.09.2026	Reviewed the work remaining after PP2. Decided to use the 2D avatar guide in the interface instead of the procedural 3D avatar, which still looked like a stick figure because MediaPipe landmarks are joint points with no body surface; recorded the 3D avatar as future work.
04.09.2026	Traced the history of the deployed model. The only model comparison in the repository selected a GRU, while the app uses a TCN, so there was no recorded justification for the change; flagged this as the top priority to resolve.
05.09.2026	Traced the history of the deployed model. The only model comparison in the repository selected a GRU, while the app uses a TCN, so there was no recorded justification for the change; flagged this as the top priority to resolve. Located the PP2 training notebook and reconstructed the full model-selection history: the parameter-matched comparison (BiLSTM-Attention 0.599, TCN 0.640, SPOTER-style 0.689 macro-F1) and the Phase 1 improvements (augmentation, velocity and acceleration features, focal loss).

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06.09.2026	Found that the DTW threshold table used by the app had been calibrated with a different distance scale from the one used for grading (hundreds versus about 2–8), so quality tiers were not meaningful. Identified the corrected v3 thresholds as the ones to use.
07.09.2026	Measured how well DTW separates different signs: about 49–51% overlap between same-sign and different-sign distances after correction, with some pairs (come, call, good, happy, walk) hard to separate. Recorded this as a limitation of movement grading.
08.09.2026	I reviewed per-sign results and run-to-run variation of the TCN (four runs, mean macro-F1 0.735 ± 0.044). Listed strong signs (milk, scared, sorry, thirsty, water) and weak ones (hurt, weak, sleep, me) as known failure modes.
09.09.2026	I manually reviewed a selection of the highest-confidence samples identified by the automated analysis. Two different movement patterns were observed: genuine concatenated signs, such as "eat" and "sit", and continuous repeated movements, such as "walk". This showed that not every repeated movement should automatically be treated as corrupted data.
12.09.2026	Traced one webcam attempt through the whole backend and found 9–11 sequential database calls, a new MediaPipe process per attempt and blocking calls on the server, which limit speed and concurrency. Documented these for later optimization. Reviewed the Parkinson's symptom education feature: confirmed which quiz version the interface uses and removed confusion caused by unused older code. Checked the literature on Parkinson's datasets, which supports keeping this feature educational only, since very few public datasets contain video.
13.09.2026	Updated the component summary document so it matches the working system (mastery data in Supabase, frontend completed, deployment approach decided), replacing outdated statements.
14.09.2026	Started drafting the research paper for SLIIT ICSCDS 2026 with the corrected literature review on sign language recognition and learning systems, ending each part with a clear research gap.

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Supervisor's Comments

Student progressing well in research project: Yes / No
If No:

Project work to be carried out before next evaluation period:

Code base checked: Yes/No Signature:

Date: 18.09.2026



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26 REGULAR RP

Research Project Periodic Log Entry Form

Project ID: R26-DS-010

Student ID: IT22178640

Name: Dayana Priyadharshani Kumar

Individual Component: AI Assisted Caregiver Workforce Deterioration Detection and Adaptive Workload Redistribution

Summary of RP work carried out during the reporting period

Date	RP Work
16.09.2026	Updated the ICSCDS (*4th International Conference on Sustainable Computing and Data Communication Systems*) manuscript to include the required author affiliations and email addresses. Sent the revised manuscript to the conference organizers and requested clarification on whether an additional upload through the submission portal was necessary.
17.09.2026	Reviewed the conference paper against the requested revisions and checked the consistency of the author block, affiliations and manuscript formatting. Continued organizing the PULSE results and supporting evaluation evidence for the reviewer-response process.
18.09.2026	Worked through the reviewer feedback relevant to PULSE. Revisited the participant-level split, uncertainty reporting and repeated participant-grouped cross-validation so that the evaluation procedure could be described more clearly in the revised paper.
19.09.2026	Rechecked the held-out evaluation results and the relationship between the main reported ROC AUC and the supporting confidence intervals. Ensured the manuscript did not confuse saved notebook results with new independent replication or live care-home validation.
21.09.2026	Revised the discussion of the PULSE baseline and ablation experiments. Explained the role of the previous stress response and the finding that the full feature set did not establish a clear improvement beyond that single baseline.
22.09.2026	Reviewed the sections covering proximity centrality and the caregiver workload recommendation feature. Clarified that proximity did not add detectable predictive value and that workload redistribution remained a human-reviewed recommendation rather than an evaluated intervention.
23.09.2026	Worked with the group to align the SENTRY, SCRIBE and GLOSS evaluation descriptions with their actual test sets and reported metrics. Checked that the common manuscript did not imply that all four components were evaluated through the same method.

24.09.2026	Prepared the point-by-point response to the conference reviewers. Matched the revised manuscript sections to the reviewer comments and described the changes without introducing unsupported performance claims.
25.09.2026	Completed another consistency review of the revised ICSCDS manuscript, including the IEEE format, page limit, author information, evaluation wording and limitations. Cross-checked the PULSE result statements against the research notebook.
26.09.2026	Reviewed the revised paper and reviewer-response letter together. Checked that every reviewer concern had a corresponding explanation and that the response identified where the relevant change appeared in the manuscript.
28.09.2026	Submitted the revised ICSCDS paper and a point-by-point response to reviewers. The revised manuscript used the IEEE conference template and included clarified PULSE evaluation procedures, participant-grouped validation and uncertainty reporting.
29.09.2026	Organized the revised conference submission materials and began consolidating evidence needed for the individual thesis. Reviewed how the PULSE methodology, model results and implementation should be documented as separate parts of the research.
30.09.2026	Continued preparing the individual and common research reports. Mapped PULSE-specific evidence to the individual thesis and identified shared architecture, integration and evaluation information for the group report.

Supervisor's Comments

Student progressing well in research project: Yes / No
If No:

Project work to be carried out before next evaluation period:

Code base checked: Yes/No Signature:

Date: 04.10.2026



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26 REGULAR RP

Research Project Periodic Log Entry Form

Project ID: R26-DS-010

Student ID: IT22057488

Name: Harishalinee Elangovan

Individual Component: Edge-Enabled Skeletal Movement Analysis System for Real-Time Patient Observable Fall Risk Detection and Prediction

Summary of RP work carried out during the reporting period

Date	RP Work
16.09.2026	Wrote thesis Chapter 7 (Discussion): analysis of domain gap (NTU subject-disjoint AUC 0.9836 vs webcam AUC 0.8527 after fine-tuning on 93 clips). Compared with NTU→UR direct-transfer result (AUC 0.410) to quantify fine-tuning benefit. Addressed PP2 environmental robustness comment via Z-drop augmentation, clip-level normalisation, and fine-tuning pipeline — all already implemented.
17.09.2026	Wrote thesis Chapter 7 limitations section: two-subject webcam dataset (30 fall / 63 normal clips); ethics form submitted but no institutional reply received; no authentication middleware on any backend endpoint; zone modifier computed but removed from scoring path after posture jitter caused false HIGH states in integration testing. Future work: multi-site validation, backend authentication layer, proper IRB approval.
18.09.2026	Wrote thesis Chapter 8 (Conclusion) and Acknowledgements. Restated key contributions: edge-deployable ST-GCN+Fusion, 18-feature biomechanical set, SITTING safety override, contextual post-processing, web+mobile caregiver system. Acknowledgements stated: "This research received no external funding." Supervisor and group members acknowledged by name.
19.09.2026	Wrote thesis Appendix A (Table A.2): five discrepancies between submitted conference paper and thesis. Paper errors: dataset size (4,797 → 3,847), webcam calibration claim, baseline split labels, fine-tuning scope description ("last two layers" → "ST-GCN auxiliary head and fusion head"), posture state count (5 → 6). Thesis values confirmed correct for each item.



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20.09.2026	Ran complete pytest test suite: pytest tests/ -v. All 49 tests passed — test_api (17), test_features (8), test_models (13), test_postprocessor (11). Saved terminal output as evidence for Appendix B Table B.2 automated-test verification row. This run was personally executed, not taken from prior project record.
21.09.2026	Performed full THESIS_FACT_VERIFICATION pass: checked every numerical claim in thesis against source files (eval_report.json, eval_report_subj.json, rf_baseline_results.json, latency_benchmark.json, split_idx.npz, features.py, data_processor.py, server.py, finetune_webcam.py). All 50 verification items confirmed correct. RF random-split confusion matrix [[426,5],[7,140]] verified to sum to 578 and included in Table 6.1.
22.09.2026	Corrected research paper (Component2_IEEE_Paper_v6.docx) for the five discrepancies identified in thesis Appendix A: updated dataset size to 3,847, removed unsupported webcam calibration claim, corrected split labels, updated fine-tuning scope description, updated posture state count to six. Paper corrections applied before viva so both documents are consistent.
23.09.2026	Thesis formatting pass: consistent heading hierarchy, figure captions numbered sequentially, table captions placed above tables, page numbers verified, Table of Contents updated to final page numbers, List of Figures and List of Tables checked against actual content. Word count confirmed within institutional limit.
24.09.2026	Thesis proofreading pass: all abbreviations defined on first use, all figure and table cross-references correct. F17 description confirmed as skeleton bounding-box height ratio (min_hip_height_ratio misnomer noted in text). Ethics statement confirmed: form submitted, no institutional reply, no approval claimed. Fine-tuning table confirmed: both ST-GCN auxiliary head and fusion head trained epochs 1–3, full unfreeze epoch 4+.
25.09.2026	Contributed SENTRY section to group project website. Wrote system overview, included architecture diagram (camera → MediaPipe → dual-branch model → post-processing → web + mobile), results table (subject-disjoint AUC 0.9836, latency 5.55 ms), and mobile app screenshots. Confirmed group website deployed and accessible.
26.09.2026	Prepared individual viva defence slides: problem and gap (3 slides), system architecture (4 slides), dataset and training (3 slides), all three evaluation protocol results (4 slides), elder home qualitative results (2 slides), live demo walkthrough



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	plan (2 slides), limitations and future work (2 slides). Total 20 slides within time allocation.
27.09.2026	Rehearsed viva response to PP2 environmental robustness comment. Three-point answer: (1) Z-drop augmentation trains robustness to missing/noisy depth signal; (2) clip-level joint normalisation removes camera-height bias so model generalises across room setups; (3) fine-tuning pipeline provides structured domain adaptation when deployed in a new environment. No additional code changes needed.
28.09.2026	Conducted mock viva with two group members as examiners. Questions addressed: F17 definition and min_hip_height_ratio misnomer; SITTING override logic; zone modifier bypass rationale; webcam evaluation threshold-selection methodology (AUC is primary threshold-independent metric; threshold 0.32 selected on Subject 2 evaluation set — noted as limitation); RF random-split reinstatement rationale.
29.09.2026	Revised thesis based on mock viva feedback: added one clarifying sentence in Section 4.4.5 explaining that the SITTING cap operates at the post-processing layer and does not affect the model-level recall reported in Tables 6.1 and 6.2. Regenerated PDF. No structural changes required. All other queried items were already clearly stated.
30.09.2026	Submitted final individual thesis draft to supervisor for review. Prepared submission package: thesis PDF, source code ZIP (skeleton coordinates only, no raw video per privacy commitment), corrected research paper PDF, bi-weekly log bundle. All Phase 11 thesis tasks marked DONE in Component Completion tracker.

Supervisor's Comments

Student progressing well in research project: Yes / No
If No:

Project work to be carried out before next evaluation period:

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Code base checked: Yes/No

Signature:

Date: 04.10.2026



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26 REGULAR RP

Research Project Periodic Log Entry Form

Project ID: R26-DS-010

Student ID: IT22294098

Name: Vidursha Prabagaran

Individual Component: Context-Aware Intelligent System for Automated ADL Extraction and Real-Time Documentation from Caregiver Voice Logs

Summary of RP work carried out during the reporting period

Date	RP Work
17.09.2026	I added a protected vocabulary of medication and food names to the grammar correction step, so that names such as Metformin, Losartan and Kiribath are not changed during correction. I retested the step with sample sentences to confirm that the protected names stayed unchanged.
18.09.2026	I recorded a 51.0-second test recording containing nine observations to test the complete updated pipeline. The old pipeline transcribed only 42 words, while the new pipeline transcribed all 71 words and produced nine records, eight of which had the correct category.
19.09.2026	I analysed the alert errors on the 200-case safety test set and found that most missed alerts were cases where T5-small extracted the correct fields but set the alert flag wrongly. Based on this finding, I designed field-based alert rules that decide the alert from the extracted fields.
20.09.2026	I evaluated the field-based alert rules on the 200-case safety set. Alert recall improved from 82.0% to 93.3% and precision from 96.1% to 100%, and all false alarms were removed.
26.09.2026	I analysed the alerts that were still missed after applying the rules. They were caused by empty symptom names, a reversed meal intake and a sentence without an alert keyword. I documented these as limitations of the extraction model for the thesis.

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27.09.2026	I measured the end-to-end processing time of the complete pipeline on a Google Colab T4 GPU. The 51-second recording took 12.2 seconds on average over three runs, which is faster than real time.
28.09.2026	I prepared a CPU timing notebook to measure the processing time of each pipeline stage (Whisper, BART-base, LanguageTool and T5-small) on a CPU runtime.
30.09.2026	I measured the processing time on a Colab CPU runtime, where the 51-second recording took 77.2 seconds on average. I confirmed that the CPU produced exactly the same records and alerts as the GPU, and found that speech recognition took the largest share of the processing time.

Supervisor's Comments

Student progressing well in research project: Yes / No
If No:

Project work to be carried out before next evaluation period:

Code base checked: Yes/No
Signature:
Date: 04.10.2026



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26 REGULAR RP

Research Project Periodic Log Entry Form

Project ID: R26-DS-010

Student ID: IT22289384

Name: Kaushalya Nagenthraraja

Individual Component: Personalized Sign Language Tutoring System for Caregivers

Summary of RP work carried out during the reporting period

Date	RP Work
16.09.2026	Continued the literature review with movement comparison (DTW) for feedback and the use of language models to phrase feedback, and wrote the gap statements that position GLOSS.
17.09.2026	Prepared a summary of the user requirements gathered from the caregiver training-school interview and the elder-care industry visit, including the recommendation that led to the Parkinson's education feature.
20.09.2026	Reorganised the research code into three Colab notebooks (01 data preparation, 02 model training and selection, 03 feedback system) and created one evidence folder so every reported number traces to a saved file.
21.09.2026	Re-ran the data preparation notebook from the raw videos to confirm the dataset is reproducible: 2,364 sequences, 59 signs and the same 70/15/15 split (355 test samples).



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23.09.2026	Retrained the three baseline models on position features and recorded their test macro-F1 (BiLSTM-Attention 0.5994, TCN 0.6397, SPOTER-style 0.6890). Trained the improved models with 441 features, augmentation and focal loss (TCN 0.6622, SPOTER-style 0.7376). Repeating the TCN training gave results between about 0.66 and 0.78, showing a single run is unstable.
26.09.2026	Fixed the model-selection method: the earlier TCN had been chosen by test score, so I applied one rule, best validation macro-F1. This chose the ensemble of the improved TCN and SPOTER-style (validation 0.8116, test 0.7752, accuracy 0.7859).
27.09.2026	Retrained the ensemble from scratch to confirm it reproduces (0.778 against 0.775) and exported it as a single model file for the app, verified to give the same outputs. Ran a bootstrap analysis (2,000 resamples): 95% CI 0.712–0.802 for the ensemble, and a gain of +0.039 (95% CI 0.008–0.072) over the best single model, which statistically supports the choice.
28.09.2026	Started the draft thesis using the department's individual thesis template: set up the chapter structure and began the introduction and background chapters. Analysed the ensemble's confidence: at 0.6 or above it is correct 98.4% of the time, and it asks learners to repeat less often than the single TCN (22.3% vs 30.4%). Measured inference speed: 52–97 ms per prediction.
29.09.2026	Compared the old and corrected DTW thresholds on the same attempts: the old ones graded all 12 as "strong", while the corrected v3 thresholds gave 2 strong, 4 moderate and 6 weak, so grading becomes informative. Audited the Gemini-reworded feedback for all 59 signs. A keyword checker passed 54/59, but a manual audit found 10 serious problems (unconfirmed sign names, dropped uncertainty, "weak" misread). Traced the cause to the prompt.
30.09.2026	Redesigned feedback as a guarded pipeline: no Gemini below 0.35 confidence, Gemini sees only the rule-based text, and every reply is checked with fallback to the rule text. The rerun removed all 10 serious problems.

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Supervisor's Comments

Student progressing well in research project: Yes / No
If No:

Project work to be carried out before next evaluation period:

Code base checked: Yes/No

Signature:

Date: 04.10.2026



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26 REGULAR RP

Research Project Periodic Log Entry Form

Project ID: R26-DS-010

Student ID: IT22178640

Name: Dayana Priyadharshani Kumar

Individual Component: AI Assisted Caregiver Workforce Deterioration Detection and Adaptive Workload Redistribution

Summary of RP work carried out during the reporting period

Date	RP Work
01.10.2026	Worked on the individual PULSE thesis structure, including the research problem, gaps, objectives and scope. Separated the proposed aims from what the completed experiments and application actually demonstrate.
02.10.2026	Documented the PULSE methodology using the TILES-2018 data, participant-grouped separation, feature reconstruction and the two-stage LightGBM pipeline. Reviewed the timing of previous EMA responses and wearable-derived features.
03.10.2026	Prepared the results and validation sections, including the held-out test ROC AUC, confidence interval, repeated grouped validation and comparisons with the previous-stress baseline. Documented the negative proximity-centrality finding.
04.10.2026	Reviewed the limitations and discussion sections of the individual thesis. Explained the dependence on the previous EMA response, the absence of a validated long-horizon forecast and the difference between a demonstration and an actual workforce intervention.
05.10.2026	Documented the PULSE backend, frontend and data workflows for the thesis. Covered risk scoring, SHAP factors, caregiver trends, supervisor access, demonstration uploads and workload redistribution recommendations.
06.10.2026	Worked with the group on the common integrated-solution report. Reviewed the CareSense architecture, shared authentication, role-based access, Supabase data model and how the four modules operate within the same application.
07.10.2026	Consolidated screenshots and technical evidence for the individual and group theses. Checked that the figures and captions represented implemented screens and that research outcomes were not confused with application-level demonstrations.

08.10.2026	Reviewed the final individual thesis draft for consistency between methodology, results, implementation and limitations. Checked terminology for elevated stress, learned baseline and human-reviewed workload recommendations.
09.10.2026	Continued checking the research handoff and thesis evidence, including model artifact versions, the notebook outputs and the distinction between the research model and earlier application artifacts. Organized the material needed for final submission.
10.10.2026	Completed the individual thesis draft and reviewed the no-EMA investigation record, including the V1–V5 experiments. Recorded that the enhanced no-EMA model was not adopted and that the results did not justify claiming reliable survey-free stress prediction.
11.10.2026	Submitted the individual PULSE thesis and the common group thesis. Also submitted the research project portfolio website, which presents the research documentation, methodology, findings and project information rather than serving as the CareSense application.
12.10.2026	Reviewed the submitted thesis and portfolio materials and organized the supporting technical documentation. Continued working on research reproducibility, data handling and the evidence required for final project assessment.
13.10.2026	Prepared the MLOps and data ethics documentation. Reviewed the model lifecycle, reproducibility, data provenance, access controls and limitations of using licensed TILES-2018 records in the PULSE demonstration.
14.10.2026	Submitted the final checklist materials, including the MLOps document and data ethical analysis document. Continued the PULSE data-audit and forecasting-design discussion, reviewing what could be supported by the available historical measurements.
15.10.2026	Worked on deployment of the integrated research application around this date and reviewed the deployed PULSE workflows. Also continued examining the feasibility of a longer-horizon stress forecast and the need for comparisons with a simple personal-history baseline.
16.10.2026	Continued the forecasting feasibility investigation, focusing on whether the available historical data could support predictions beyond the next EMA response. Reviewed possible targets, input availability and evaluation designs, while keeping proposed approaches separate from implemented and validated results.
17.10.2026	Reviewed the PULSE technical handoff and database audit, including backend and frontend implementation, database migrations, model artifacts and inference workflows. The audit notebook had been executed through Cell 32; later cells remained unexecuted. Documented unresolved consistency and data-lifecycle issues rather than treating them as fixed.

18.10.2026	Updated the PULSE application to use the research-paper model artifacts and rescored stored results. Added Model Check and Forecast views and explanation cards across caregiver pages. Reviewed the Forecast V0 feasibility output and outstanding model-registry discrepancies. Planned a visit to Love and Care Elder Home today for user acceptance testing;

Supervisor's Comments

Student progressing well in research project: Yes / No
If No:

Project work to be carried out before next evaluation period:

Code base checked: Yes/No

Signature:

Date: 19.10.2026



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26 REGULAR RP

Research Project Periodic Log Entry Form

Project ID: R26-DS-010

Student ID: IT22057488

Name: Harishalinee Elangovan

Individual Component: Edge-Enabled Skeletal Movement Analysis System for Real-Time Patient Observable Fall Risk Detection and Prediction

Summary of RP work carried out during the reporting period

Date	RP Work
01.10.2026	Addressed review comments on thesis draft. Corrections: added missing reference for MediaPipe Pose (Bazarevsky et al., 2020), corrected one figure caption number. No structural changes required. Regenerated thesis PDF and re-confirmed Table of Contents page numbers.
02.10.2026	Completed group common thesis Chapter 6 integration section: described SENTRY–SCRIBE–GLOSS data flows through Supabase shared schema. Documented how fall_alerts table structure allows audit logging by the SCRIBE component. Confirmed no schema conflicts with other component migrations.
03.10.2026	Group common thesis proofreading pass for all SENTRY sections. Verified all cross-references to SENTRY architecture are consistent with individual thesis figures and values. Confirmed group thesis committed to repository and submitted to group supervisor for review.
04.10.2026	Final code freeze for SENTRY component. Merged SENTRY-03 PR into pp2-foundation after team review approval (backend services, web frontend, Supabase migrations). Confirmed sentry-mobile branch PR merged by team lead (Flutter caregiver app). Tagged release v1.0.0 on pp2-foundation. No further code changes planned before viva.
05.10.2026	Compiled 25 anticipated viva questions across categories: dataset design decisions, model architecture choices, post-processing parameter selection, ethical considerations (ethics form status), system limitations (no auth, two-subject webcam data), and future work. Drafted two-sentence answers for each. Timed verbal delivery of full answer set.



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06.10.2026	Second mock viva (solo using question list). Identified three answers needing tighter delivery: (1) F13 weight justification — judgement call, no grid search, stated in thesis; (2) webcam evaluation methodology — AUC is threshold-independent primary metric, threshold bias affects accuracy/recall at that operating point only; (3) zone modifier bypass — removed from scoring path after integration testing showed posture jitter caused false HIGH states.
07.10.2026	Final thesis review focused on sections most likely to draw examiner questions: Section 3.6 ethics (honest non-approval statement confirmed), Appendix A Table A.2 (five paper discrepancies with evidence confirmed), Appendix B Table B.2 (all six verification steps accurately describe what was personally executed, including pytest run on 20.09.2026). No amendments required.
08.10.2026	Prepared physical viva materials: printed thesis, printed corrected paper, printed key figures summary (eval_report.json and eval_report_subj.json). Set up laptop with live SENTRY demo: backend running, web supervisor dashboard open on Room 04 live view, Flutter mobile app connected and displaying real-time skeleton feed.
09.10.2026	Final group coordination meeting. All component PRs confirmed merged to pp2-foundation. Group website confirmed live with all three component sections complete. Common thesis confirmed submitted. Agreed group viva demo order and timing (GLOSS → SCRIBE → SENTRY). Confirmed shared Supabase project accessible from presentation venue.
10.10.2026	Final end-to-end system check: backend started, web supervisor dashboard opened, Flutter mobile app connected. Verified: live skeleton rendering at 25 FPS, metric chips updating (risk score, posture, zone, confidence), alert generation and acknowledge sync between web and mobile under 1 second, SITTING cap suppressing HIGH alerts in seated posture, replay buffer saving 90 frames on alert trigger.
11.10.2026	Submitted final thesis to SLIIT submission portal. Confirmed submission receipt email. Submitted all required supporting documents: plagiarism report, supervisor sign-off form, bi-weekly log bundle, source code ZIP. All Phase 12 submission tasks marked DONE in Component Completion tracker.
12.10.2026	Group common thesis final submission confirmed by group lead. All group members' individual thesis portal submissions verified complete. Group prepared



	combined component demo environment and backup demo video for connectivity failure contingency.
13.10.2026	Rehearsed live demo walkthrough (target: 3 minutes). Demo sequence: open web dashboard → select Room 04 → activate live feed → skeleton canvas with metric chips visible → simulate fall movement → HIGH alert generated → acknowledge on mobile → Realtime sync confirmed on web. Ran walkthrough 3 times to confirm repeatability.
14.10.2026	Reviewed Appendix A Table A.2 (five paper errors) aloud. Prepared one-sentence explanation for each: dataset size corrected after split_idx.npz count post paper-submission; webcam calibration claim was unsupported in code; baseline split labels were misstated; fine-tuning scope was mis-summarised from finetune_webcam.py; posture state count updated when UNKNOWN state was formalised in ContextEngine.
15.10.2026	Final numerical recall: 3,847 total sequences, 49 pytest tests, AUC 0.9926 (random-split fusion), AUC 0.9836 (subject-disjoint fusion), AUC 0.8527 (webcam fine-tuned), AUC 0.410 (NTU→UR direct transfer), 5.551 ms end-to-end latency, 180.1 windows/s, EMA α=0.25, τ_high=0.72, ALERT_COOLDOWN_S=20, 6 posture states, 18 features, 7 augmentations, 3 ST-GCN blocks.
16.10.2026	Confirmed laptop demo environment stable: backend, web dashboard, and Flutter mobile app all running simultaneously without crash or latency spike. Prepared backup printed figures in case of connectivity failure
17.10.2026	Identified and resolved critical false-positive HIGH risk issue: root cause was misaligned risk thresholds (RISK_TAU_LOW=0.55, RISK_TAU_HIGH=0.72) against calibrated model decision boundary (fusion_threshold=0.94). Fixed by loading thresholds dynamically from threshold.json (TAU_LOW=0.75, TAU_HIGH=0.89). Added warmup EMA reset on first full buffer fill and confirmed-STANDING posture cap at MODERATE. Verified skeleton replay showing Risk: 0.9949 on fall posture with peak vertical acceleration and kinetic

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	energy spike as top contributing features. Pushed all fixes to SENTRY-03. Added MODEL_CARD.md documentation.
18.10.2026	Final viva preparation: verified all model weights (stgcn_best.pth, fusion_best.pth, classifier_best.pth, rf_baseline.pkl, feature_scaler.joblib, threshold.json) committed and pushed to SENTRY-03 for leader integration. Confirmed no retraining required on deployment. Practiced fall demonstration sequence: standing normal → NORMAL, sudden forward drop held 3s → HIGH alert fires, skeleton replay confirms biomechanical fall signature. Generated 5 research portfolio charts (model evaluation, ablation study, risk threshold analysis, zone modifier impact, alert response time latency).

Supervisor's Comments

Student progressing well in research project: Yes / No
If No:

Project work to be carried out before next evaluation period:

Code base checked: Yes/No
Signature:
Date: 19.10.2026



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26 REGULAR RP

Research Project Periodic Log Entry Form

Project ID: R26-DS-010

Student ID: IT22294098

Name: Vidursha Prabagaran

Individual Component: Context-Aware Intelligent System for Automated ADL Extraction and Real-Time Documentation from Caregiver Voice Logs

Summary of RP work carried out during the reporting period

Date	RP Work
06.10.2026	I gathered the materials needed for my individual thesis, including the thesis template, the formatting standards, photographs of care home records with names obscured, result charts and project details. I also planned the structure of the thesis chapters.
07.10.2026	I drafted the main thesis chapters, including the introduction, literature review, research methodology, design and implementation. I described the research design, the dataset creation process and the SCRIBE pipeline in detail.
08.10.2026	I completed the remaining thesis chapters, including results, discussion, commercialisation and conclusion, together with the appendices. I added the figures and tables and reviewed the complete draft.
09.10.2026	I continued writing and improving my individual thesis.
10.10.2026	I continued writing and improving my individual thesis, and checked all chapters for consistency.



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11.10.2026	I finalised the thesis layout by adding the care home photograph, correcting the page numbers in the Table of Contents, List of Figures and List of Tables, adjusting the cover page layout, and updating the framework title in Figure 1.1. I then submitted my individual thesis draft, the group thesis and the website.
12.10.2026	After submitting the thesis and website, I continued working on the frontend and backend of the SCRIBE web application.
13.10.2026	I prepared my sections of the group MLOps report, covering the data pipeline, model development, model integration, deployment and future enhancements for the SCRIBE component. I also prepared the ethical analysis for my component, covering data privacy, data security, bias and fairness, transparency and accountability, and the impact on stakeholders.
14.10.2026	I submitted the Ethical Analysis Report and the MLOps Report. I also updated the SCRIBE web application to support long recordings and multiple observations, adding audio chunking and sentence splitting in the backend and saving one record per observation through a new SCRIBE database migration. In addition, I expanded the protected vocabulary of medication and food names, added a hybrid T5-small and rule-based extraction layer, improved the symptom category rule with a negation check, and tested the changes with sample recordings.
15.10.2026	I analysed the caregiver and supervisor dashboards and planned improvements to their design and user experience. I worked on card views for the summaries, a single Play/Stop audio button, search boxes for tables, an improved recording screen, and a colour palette that matches the shared CareSense sidebar.
16.10.2026	I improved the period summary so that it updates automatically when data changes, uses Sri Lanka time for day boundaries, includes a short spoken summary of the key points, and shows all caregivers assigned during the selected period. I also reviewed the complete pipeline and all the code of the SCRIBE component to make sure everything worked correctly.

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17.10.2026	I organised the remaining project work by adding tasks in Microsoft Teams for integration, results and analysis, testing and validation, and the final presentation and demo. I also tested the SCRIBE system with spoken sample sentences to make sure it was ready for testing at the care home.
18.10.2026	We visited the Love & Elder Care Home as a group to test the CareSense system. I tested the SCRIBE component by recording caregiver observations and checking the generated ADL records, alerts and summaries. The system worked correctly in the real care home environment, and the staff responded positively to it.

Supervisor's Comments

Student progressing well in research project: Yes / No
If No:

Project work to be carried out before next evaluation period:

Code base checked: Yes/No
Signature:
Date: 19.10.2026

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26 REGULAR RP

Research Project Periodic Log Entry Form

Project ID: R26-DS-010

Student ID: IT22289384

Name: Kaushalya Nagenthraraja

Individual Component: Personalized Sign Language Tutoring System for Caregivers

Summary of RP work carried out during the reporting period

Date	RP Work
01.10.2026	Read every Gemini-reworded feedback message by hand and found 8 that dropped important details such as "left wrist". Added a detail check (guard v2): 44 rewordings are now shown and 10 fall back to the rule-based text. Produced the thesis figures (confusion matrices, per-sign F1, model comparison, confidence bands).
03.10.2026	Wrote the introduction chapter of the draft thesis: the caregiver communication problem, the research gap, and the main and specific objectives. Explained the scope of GLOSS as teaching non-verbal care signals: signs from the person (59 ASL signs) and signs from the body (Parkinson's motor symptoms, educational only, no classifier).
04.10.2026	Wrote the commercialisation section with monthly running costs (Supabase, backend hosting, Gemini) and pricing tiers for individuals, care homes and training schools. Prepared the user evaluation kit: practice signs, a 7-question Parkinson's quiz, SUS and GLOSS-specific questionnaires.
05.10.2026	I wrote the methodology and implementation chapters: data preparation, model training and selection, DTW movement grading, guarded feedback, mastery tracking and adaptive lesson selection. Justified the reduction from 66 to 59 signs (low-sample and rarely used signs removed after checking with a sign language tutor) and the choice of ASL.



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06.10.2026	Locked all final results in one results sheet so every number in the thesis comes from one source. Prepared the model bundle for the app with file fingerprints, and wrote the results and discussion chapters (ensemble macro-F1 0.7752, confidence analysis, DTW limits, feedback faithfulness).
07.10.2026	Integrated the final ensemble model into the app and verified it against the notebook: model and full-pipeline parity 10/10, with identical grading and feedback. Completed the literature review and requirements chapters, using the caregiver training-school interview and the elder-care industry visit as user evidence. Added the limitations, risks (dataset licences, sign-language localisation) and future work (other conditions such as dementia and stroke, Sri Lankan Sign Language).
08.10.2026	Prepared the GLOSS content for the research project website: the problem, objectives, methodology, system screenshots and key results. Finalised the draft thesis: added the abstract, front matter, IEEE references and appendices (including the AI-use disclosure), applied the SLIIT formatting guide (Times New Roman, 1.5 spacing, chapter-numbered figures and tables), and checked every number against the results sheet.
09.10.2026	Submitted the draft thesis and the research project website. Prepared the items for the final checklist: code repository, model files, evidence folder, notebooks and documentation.
11.10.2026	Submitted the final checklist. Carried out a full readiness audit of GLOSS (research, interface, model, features, examiner questions), which identified the remaining gaps: the avatar, no-sign detection, response speed, learner motivation and supervisor monitoring. Started re-extracting MediaPipe landmarks from the reference videos to drive a 3D avatar.
13.10.2026	Converted the re-extracted landmarks into joint rotations for a 3D avatar. Tried external VRM avatar files, but no free model worked reliably with the converted motion, so built the avatar in code with three.js (joint limits, smoothed hand movement). Fixed the mismatch where the 2D guide signed with the right hand and the 3D avatar with the left.
14.10.2026	Improved recognition reliability. Added no-sign detection with confidence gates (below 0.35 the app shows "Not recognised" and asks for a repeat), a live hand and framing overlay with specific retry messages, and 30 fps capture. Reduced attempt time with a persistent MediaPipe worker and non-blocking Gemini calls, and fixed lost practice sessions.

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15.10.2026	Added learning features. Check my sign lets caregivers try any sign and get free recognition. Recall challenge tests signs without the guide, and a strong recall attempt 7 or more days later marks a sign as Retained. Built My Journey with levels and badges (a sign is Demonstrated after 3 strong attempts in a row) and merged History into the Progress page.
16.10.2026	Added supervisor monitoring. Both supervisors can see every caregiver's progress report, and supervisors can also learn signs themselves. Secured this with database access rules and kept test attempts out of real caregiver records. Pushed the GLOSS-02 branch with the model files to GitHub for the team leader to merge.
17.10.2026	Checked that the GLOSS-02 branch works on another machine. Cloned it fresh, installed only the listed libraries, and confirmed the model fingerprint, that all tests pass and that no secrets were committed. Wrote the setup notes and merge notes the team leader needs to merge it into the integrated platform.
18.10.2026	Reviewed every 3D avatar demonstration against its reference video, and trimmed or corrected the signs whose movement looked wrong. Started replacing the studio reference clips with the medoid clips selected from the dataset, so each demonstration matches the movement the grading compares against.

Supervisor's Comments

Student progressing well in research project: Yes / No

If No:

Project work to be carried out before next evaluation period:

Code base checked: Yes/No

Signature:

Date: 19.10.2026